

How to succeed

- Read the Chapter ahead of lectures
- Come to class
- **Start paper early**
- Study groups
- Prepare exams
- Don't cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)...**
- Ask questions and Think skeptically

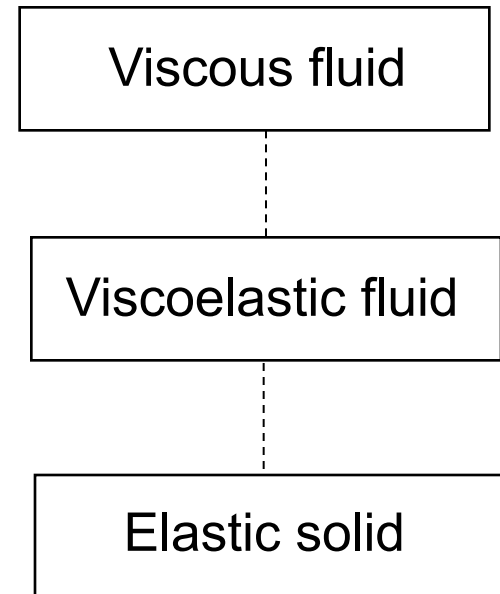
Introduction to Polymer Materials

Lecture 7: Viscoelasticity

SUN Dazhi

Introduction to Viscoelasticity

- All viscous liquids deform continuously under the influence of an applied stress – **They exhibit viscous behavior.**
- Solids deform under an applied stress, but soon reach a position of equilibrium, in which further deformation ceases. If the stress is removed they recover their original shape – **They exhibit elastic behavior.**
- **Viscoelastic fluids** can exhibit both viscosity and elasticity, depending on the conditions.



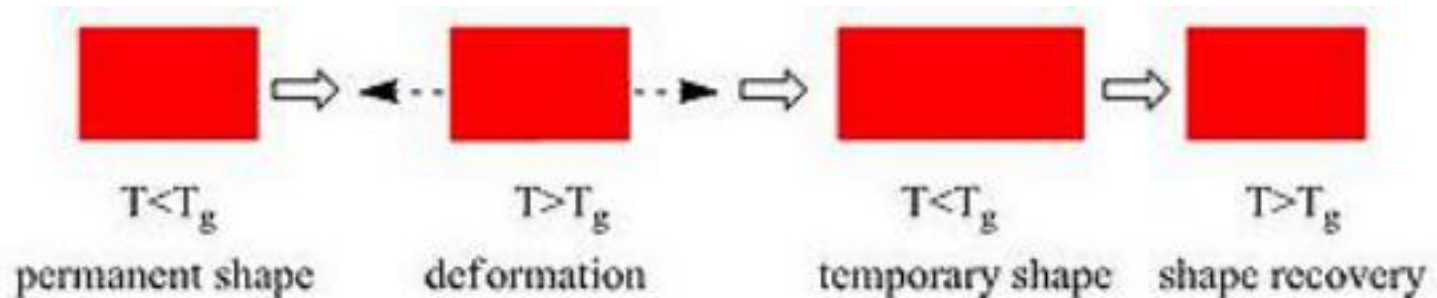
Polymers display VISCOELASTIC properties

A Demonstration of Polymer Viscoelasticity



Poly(ethylene oxide) in water

Shape Memory Polymers



Viscoelasticity

Is “silly putty” a solid or a liquid?

<http://my.tv.sohu.com/us/63326670/62648378.shtml>

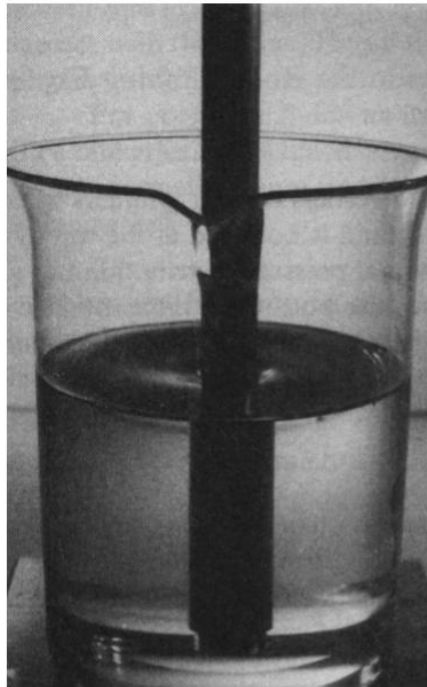
Polymers have both Viscous (liquid) and elastic (solid) characteristics

Weissenberg Effect

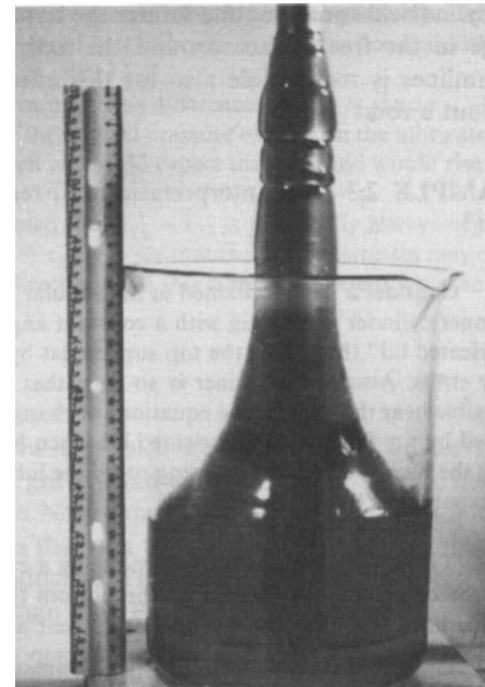
First normal stress effects: rod climbing

$$\tau_{11} - \tau_{22} < 0$$

Extra tension in the 1-direction pulls azimuthally and upward (see DPL p65).

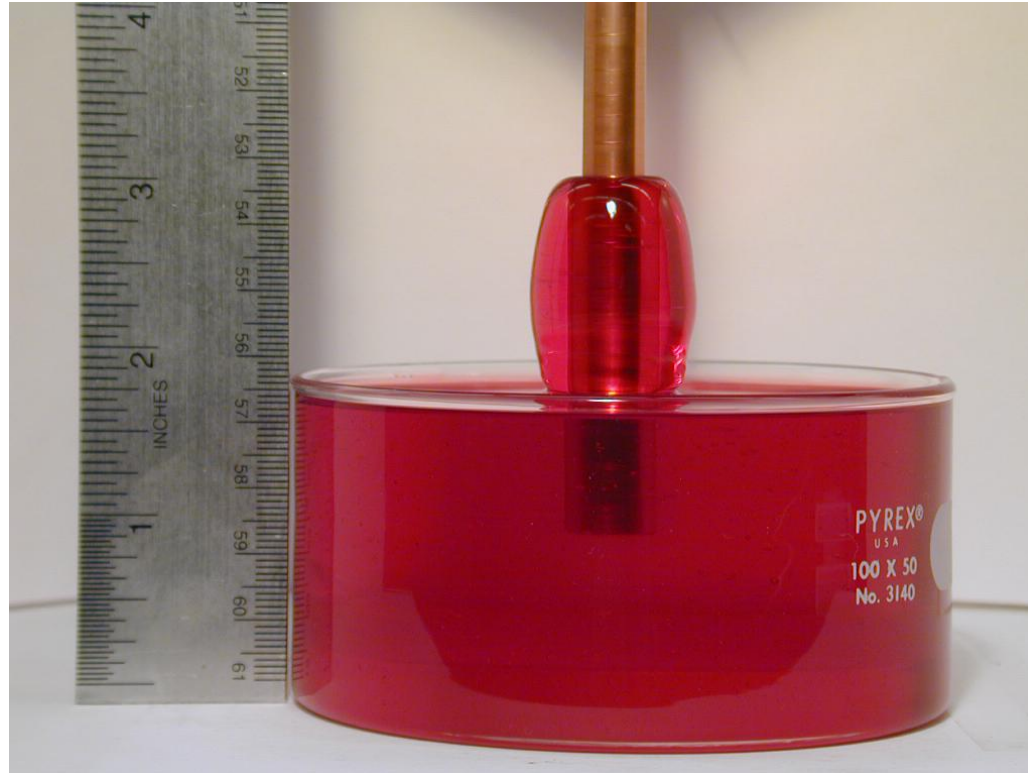


Newtonian - glycerin



Viscoelastic - solution of polyacrylamide in glycerin

Video



Dough Climbing: Weissenberg Effect



<http://tinypic.com/player.php?v=ie9mo&s=8#.VEMh7MbUOGk>

Barus Effect



Kaye Effect



What is Rheology?

Rheology is the science of flow and deformation of matter

4

Introduction. Rheology: Subject and Goals

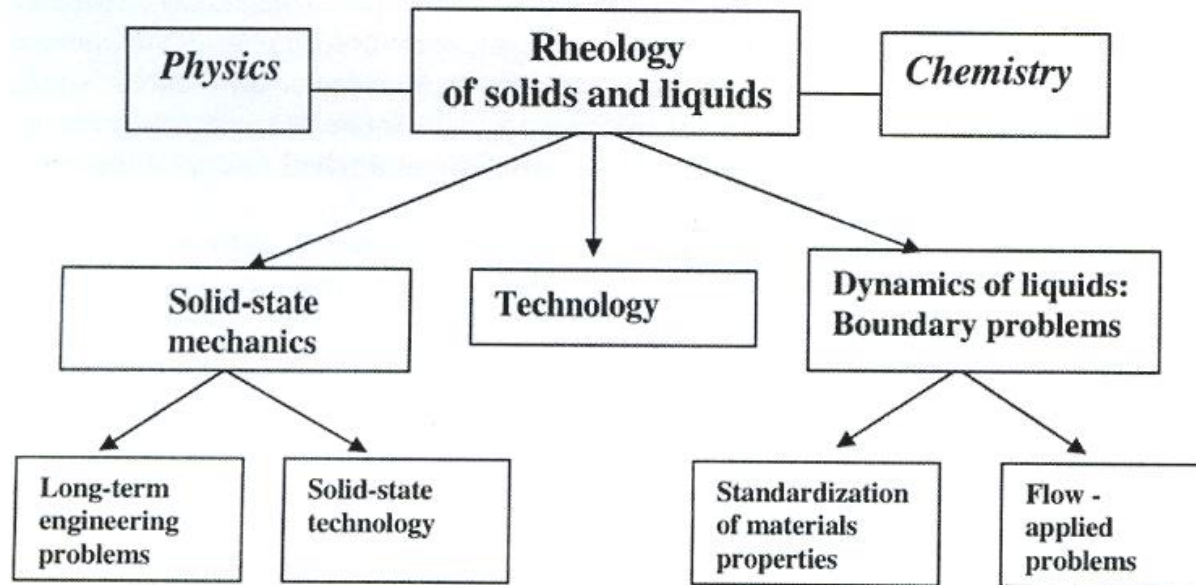
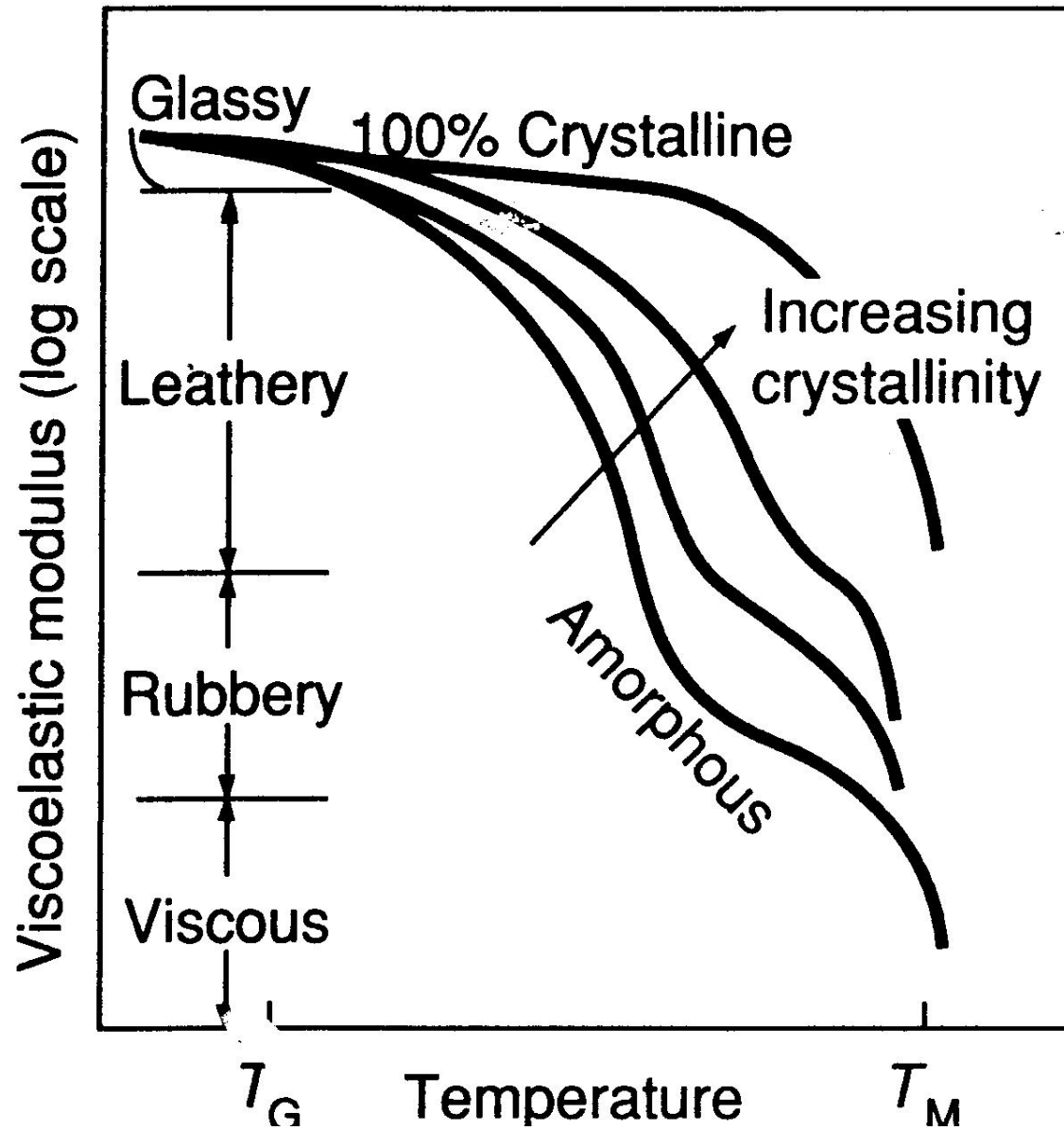
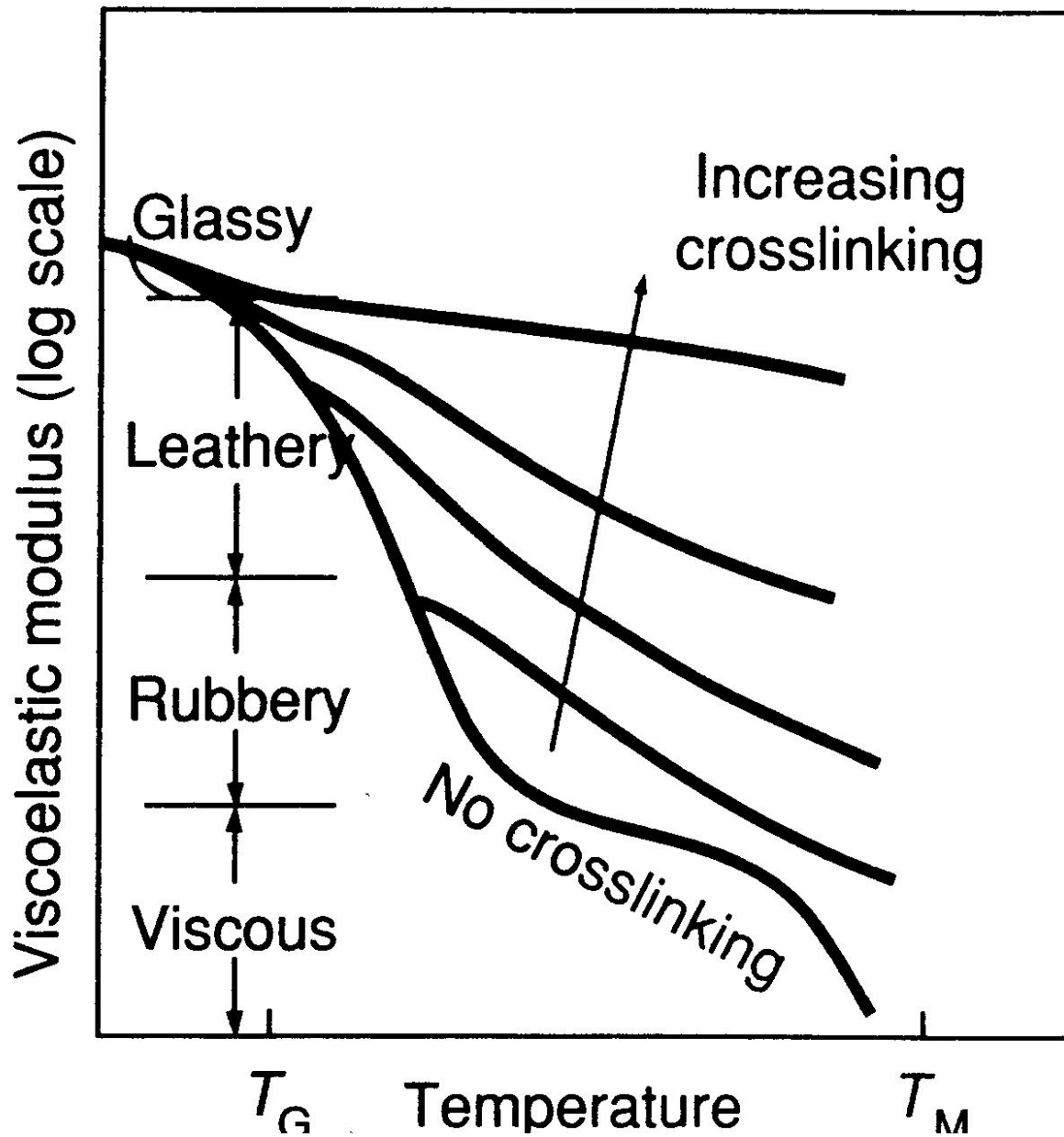


Fig. 1. Rheology as an interdisciplinary science – its place among other sciences.





Time dependent processes: Viscoelasticity

The response of polymeric liquids, such as melts and solutions, to an imposed stress may resemble the behavior of a solid or a liquid, depending on the situation.

- **Liquid favored by longer time scales & higher temperatures**
- **Solid favored by short time and lower temperature**

M. Reiner is credited with naming the Deborah Number after the song of Deborah, Judges 5:5- "The mountains flowed before the Lord" (Fig. 3.10). It was first mentioned in his article "The Deborah Number" in the January 1964 issue of Physics Today.

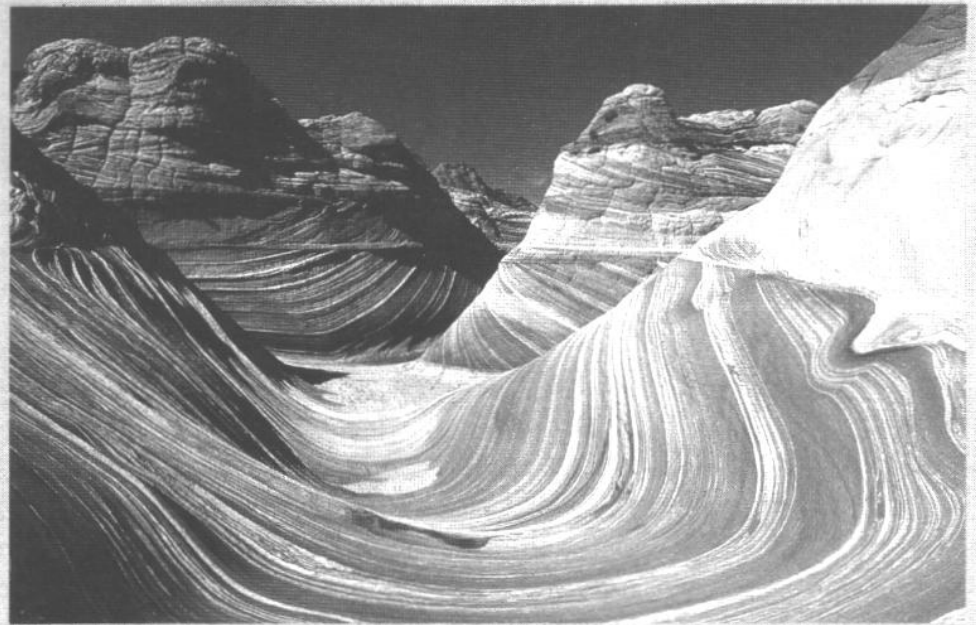


Figure. 3.10 Nestles Canyon, Arizona. Courtesy of Wolfgang Cohnen (©1997)

$$De \equiv \frac{\text{characteristic relaxation time}}{\text{time scale of the deformation}} = \frac{\lambda_c}{t_s}$$

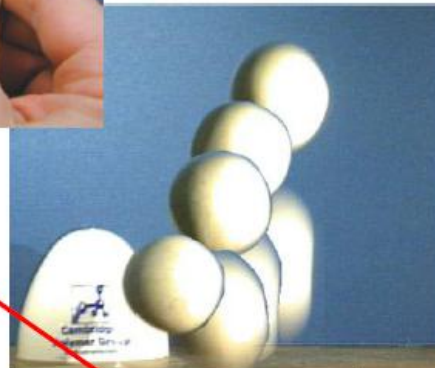
De is large: solid behavior; small: liquid behavior.



At very long times the material behaves like a liquid spreading out onto a flat surface



At moderate times the Silly Putty™ stretches like a plastic solid

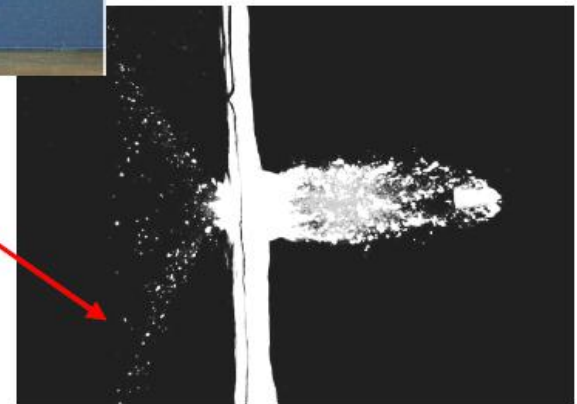


At short times the Silly Putty™ bounces like an elastic solid

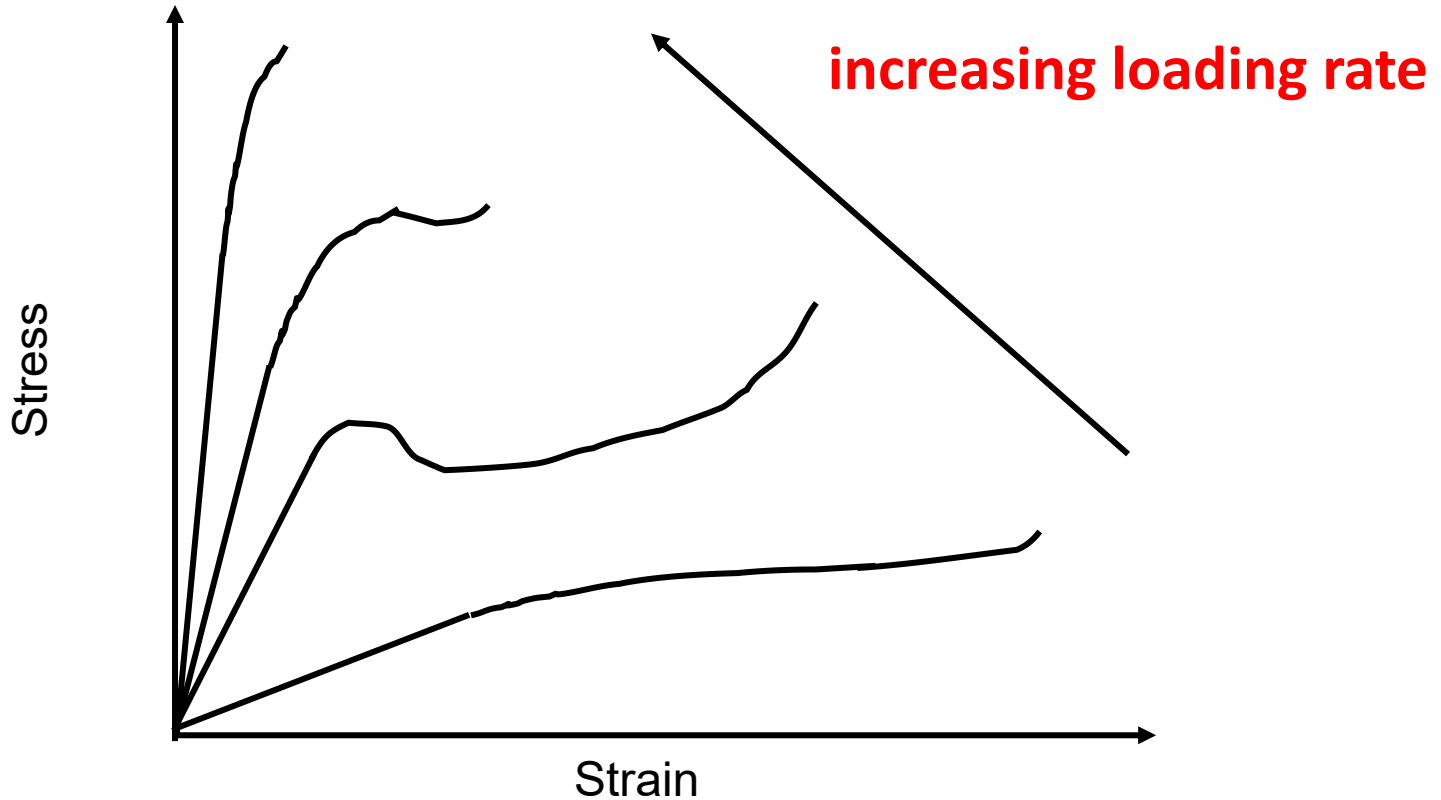
Increasing Deborah Number

$$De = \frac{\lambda_{material}}{t_{flow}}$$

At very short times (the impact of a bullet) the Silly Putty™ shatters (courtesy MIT Edgerton Strobe Laboratories).



Loading rate-dependent Mechanical Property



Time and Temperature

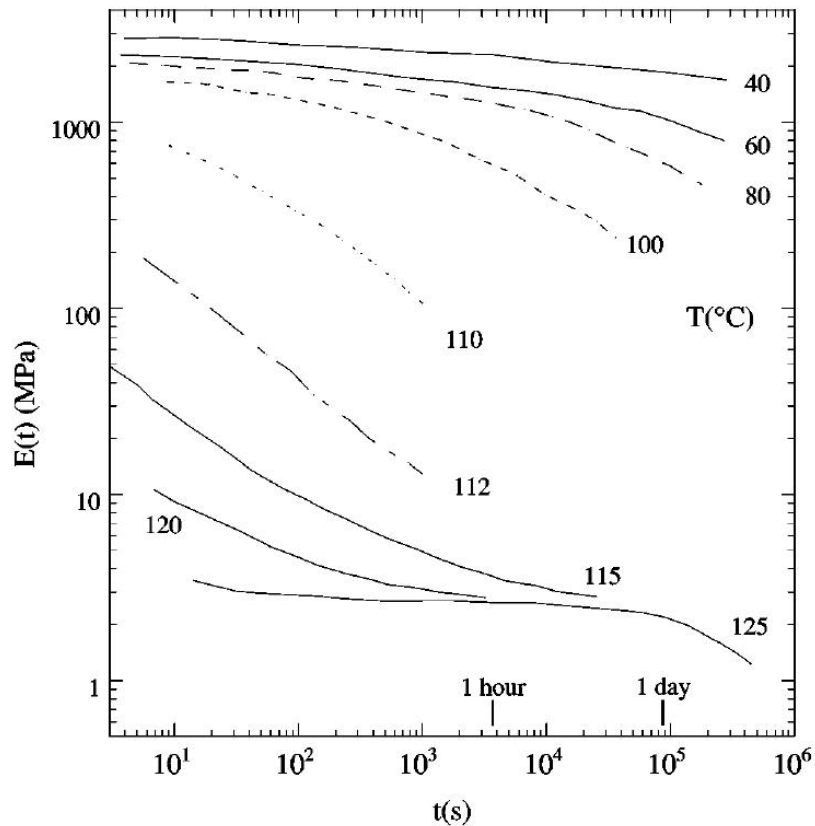


FIG. 2. Relaxation modulus $E(t)$ of PMMA at various temperatures, adapted from McLoughlin and Tobolsky.¹⁰

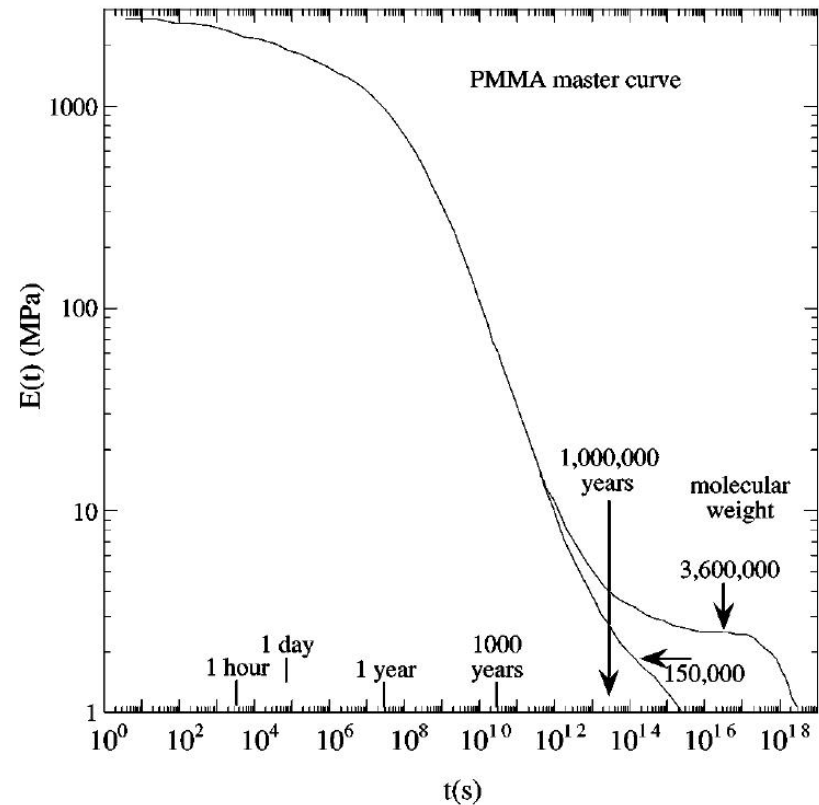
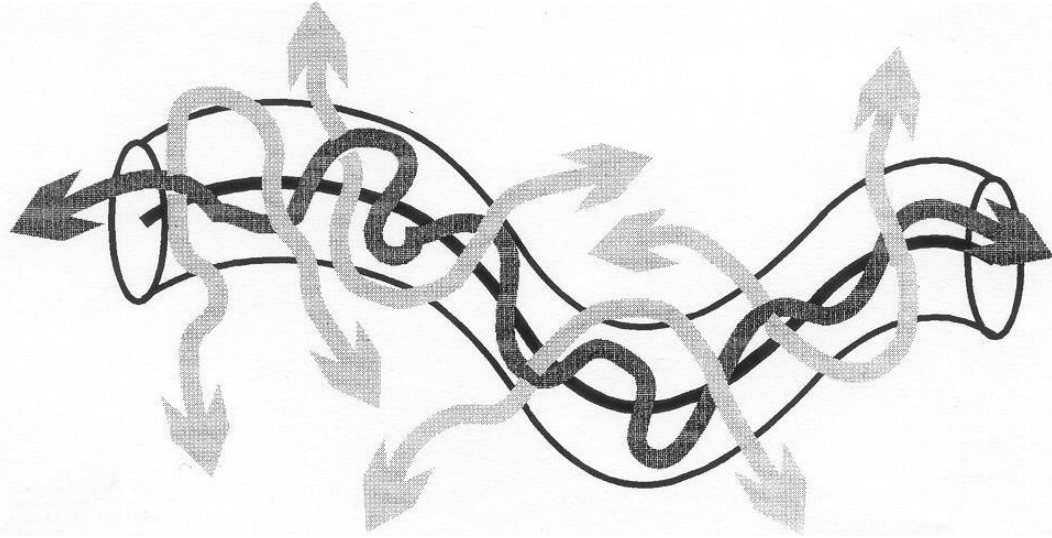


FIG. 12. Master curve for PMMA obtained from results at various temperatures via time-temperature superposition, adapted from McLoughlin and Tobolsky.¹⁰

Network of Entanglements



There is a direct analogy between chemical crosslinks in rubbers and “physical” crosslinks that are created by the entanglements.

The physical entanglements can support stress (for short periods up to a time τ_T), creating a “transient” network.

Entanglement Molecular Weights, M_e

M_e (g/mole)

Poly(ethylene)	1,250
Poly(butadiene)	1,700
Poly(vinyl acetate)	6,900
Poly(dimethyl siloxane)	8,100
Poly(styrene)	19,000

Pitch drop experiment

- Started in 1927 by University of Queensland Professor Thomas Parnell
- A drop of pitch falls every 9 years



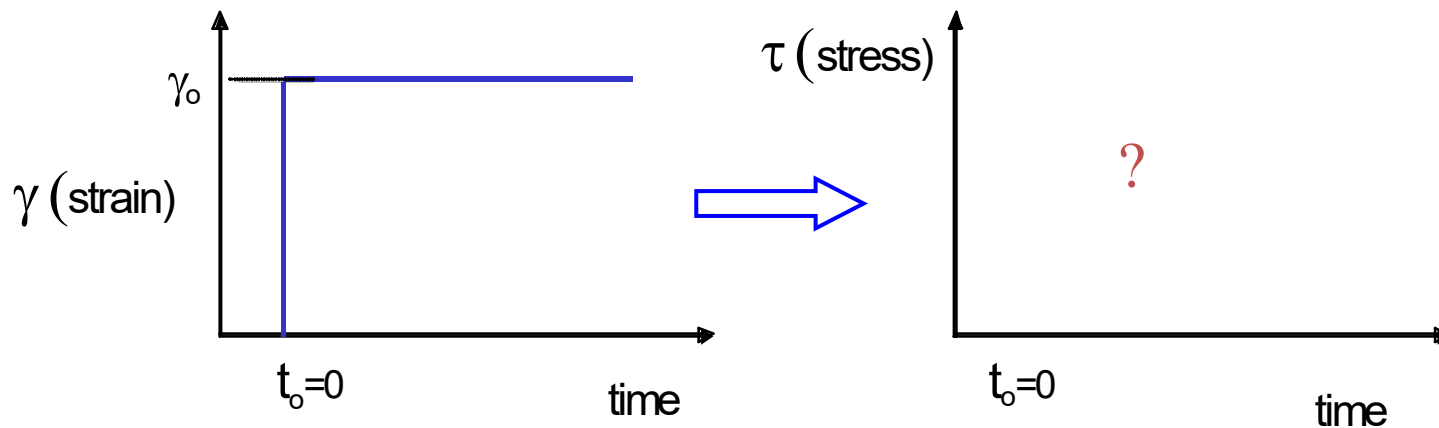
Pitch can be shattered by a hammer



Pitch drop experiment apparatus

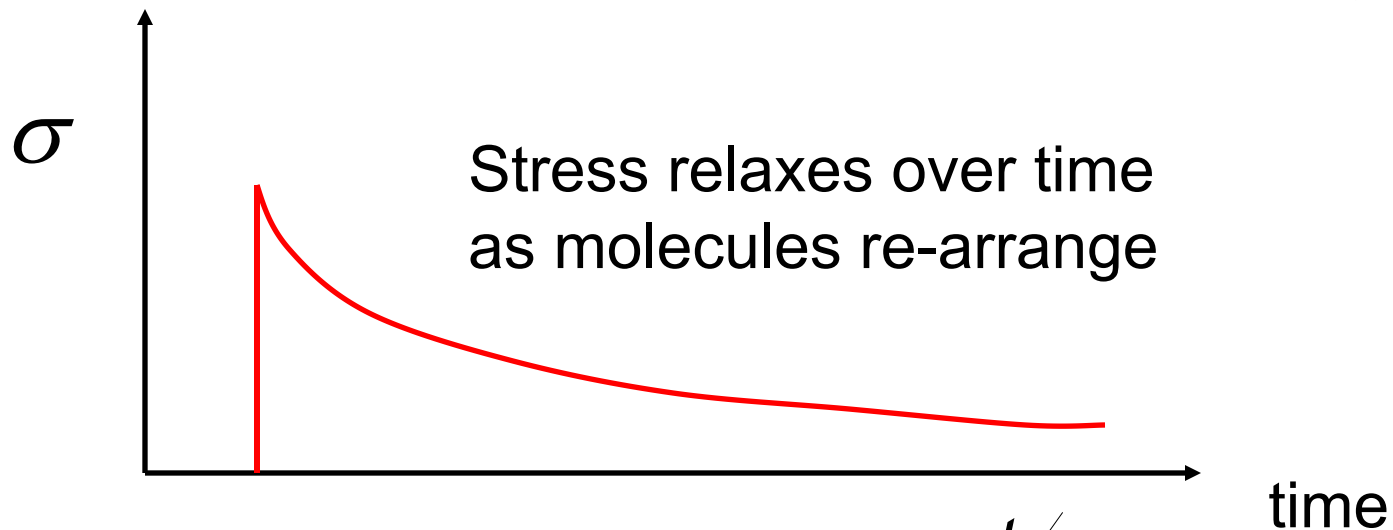
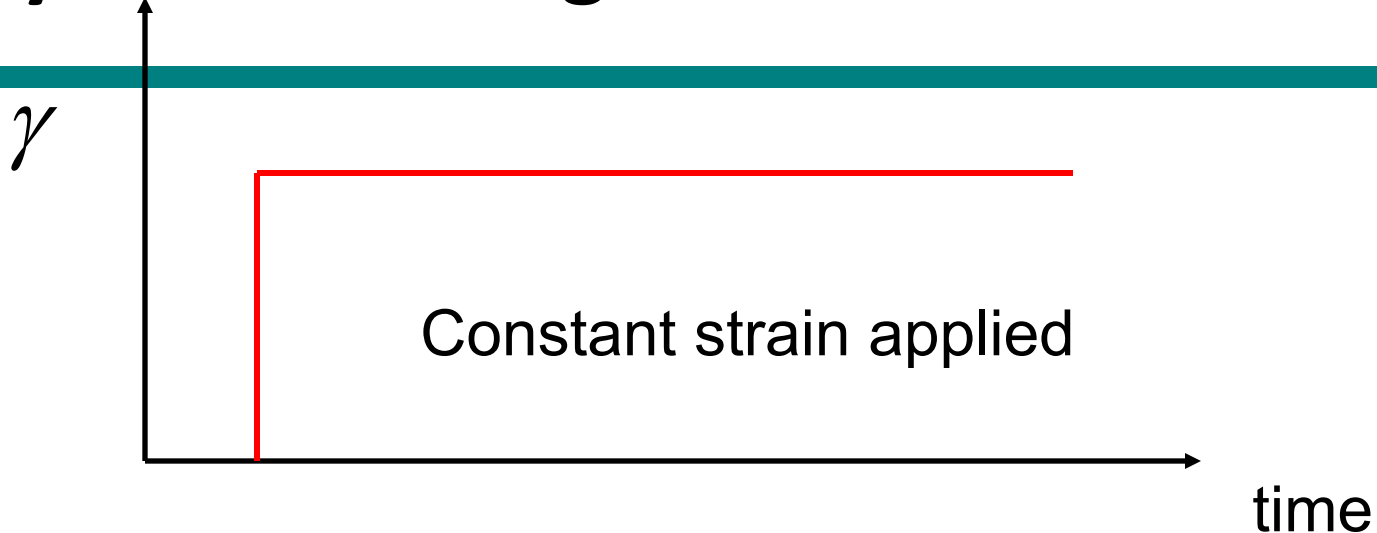
Viscoelasticity and Stress Relaxation

Whereas steady-shear measurements probe material responses under a steady-state condition, creep and stress relaxation monitor material responses as a function of time.



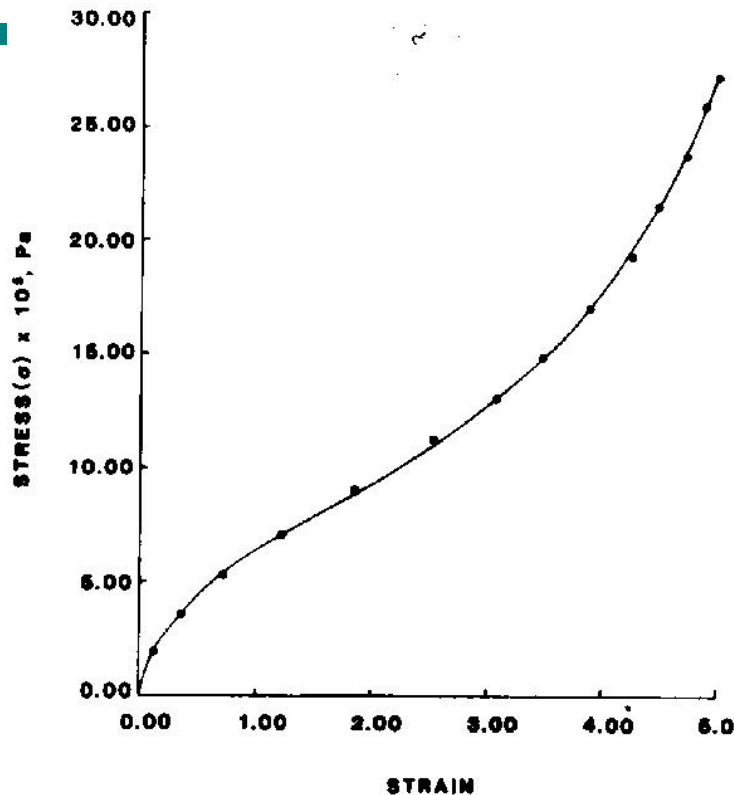
- Stress relaxation studies the effect of a step-change in strain on stress.

Physical Meaning of the Relaxation Time



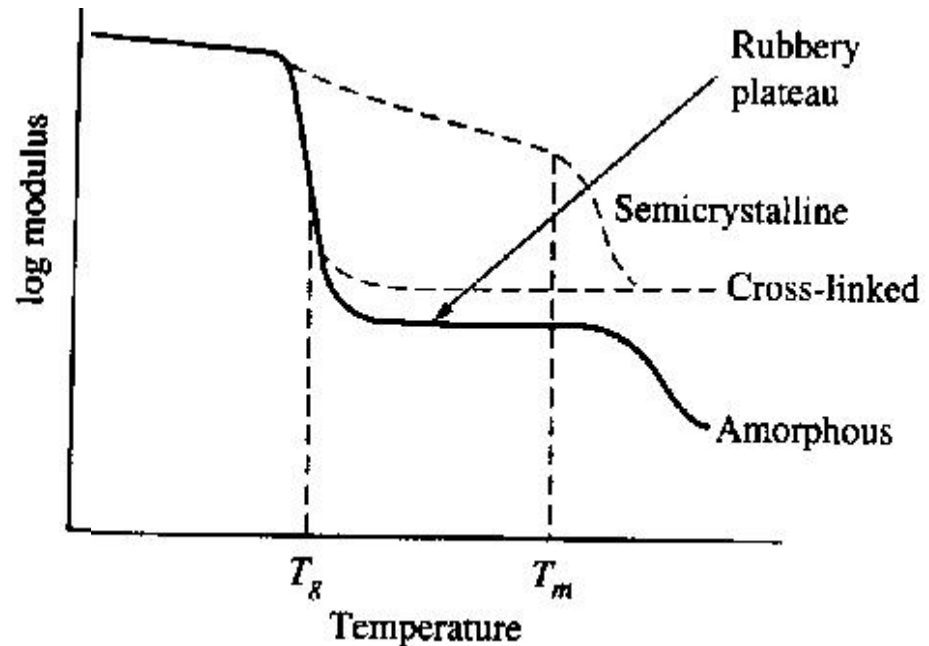
Stress relaxation: $\sigma(t) = G\gamma e^{-t/\tau}$

Static Testing of Rubber Vulcanizates

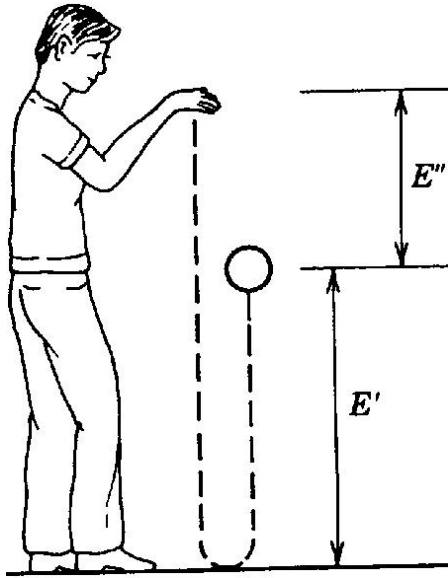


Note that at common static test conditions, vulcanized elastomers store energy efficiently, with little loss of inputted energy.

- Static tensile tests measure retractive stress at a constant elongation (strain) rate.
 - Both strain rate and temperature influence the result



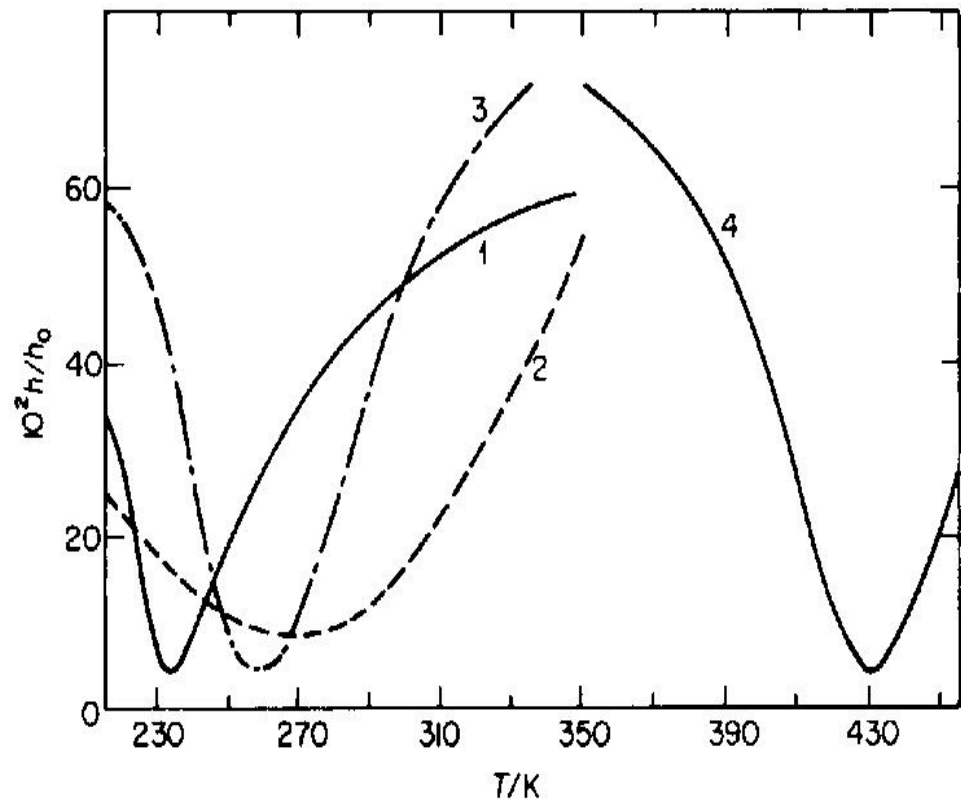
Dynamic Testing of Rubber Vulcanizates: Resilience



Change of rebound resilience (h/h_0) with temperature T for:

- 1. *cis*-poly(isoprene);
- 2. poly(isobutylene);
- 3. poly(chloroprene);
- 4. poly(methyl methacrylate).

Resilience tests reflect the ability of an elastomeric compound to store and return energy at a given frequency and temperature.



Mathematical models: Hooke and Newton

- It is difficult to predict the creep and stress relaxation for polymeric materials.
- It is easier to predict the behaviour of polymeric materials with the assumption → it behaves as linear viscoelastic behaviour.
- Deformation of polymeric materials can be divided to two components:
 - Elastic component – Hooke's law
 - Viscous component – Newton's law
- Deformation of polymeric materials → combination of Hooke's law and Newton's law.

Hooke's law & Newton's Law

- The behaviour of linear elastic were given by Hooke's law:

$$\sigma = Ee$$

or

$$\frac{d\sigma}{dt} = E \frac{de}{dt}$$

- The behaviour of linear viscous were given by Newton's Law:

$$\sigma = \eta \frac{de}{dt}$$

E = Elastic modulus

σ = Stress

e = strain

de/dt = strain rate

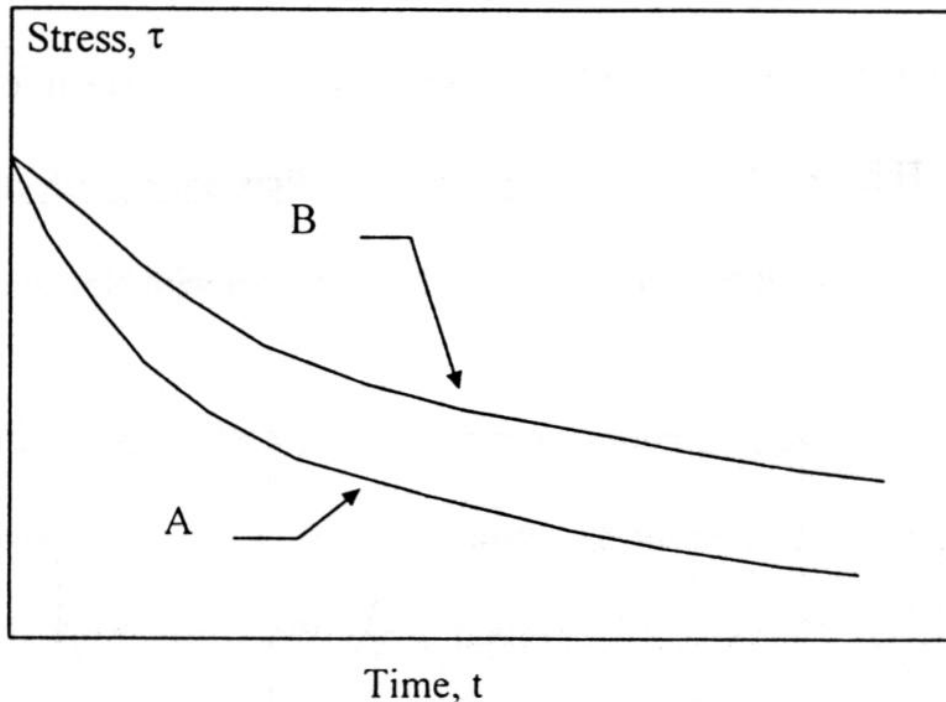
$d\sigma/dt$ = stress rate

η = viscosity

**** This equation only applicable at low strain**

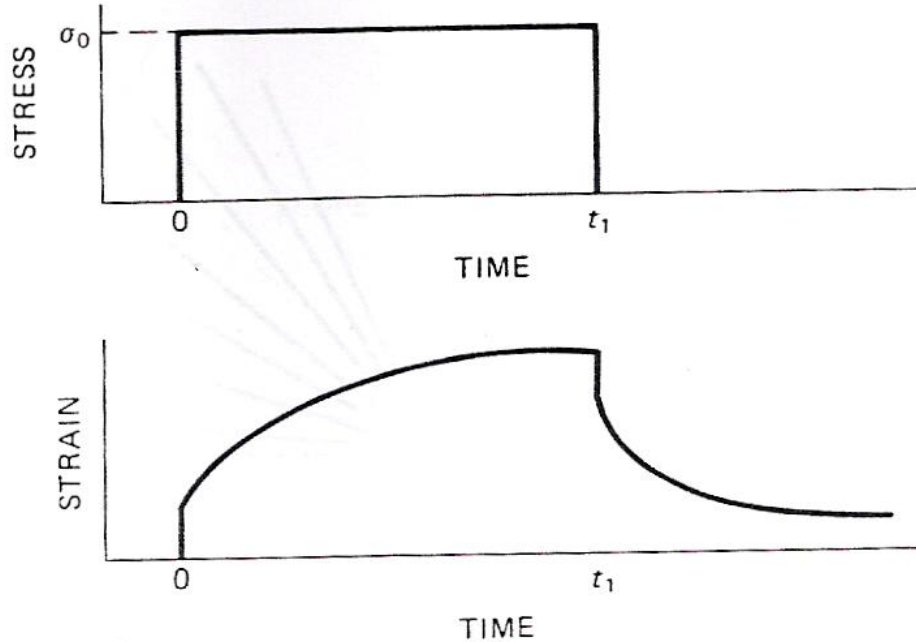
Viscoelasticity and Stress Relaxation

Stress relaxation can be measured by shearing the polymer melt in a viscometer (for example cone-and-plate or parallel plate). If the rotation is suddenly stopped, ie. $\dot{\gamma}=0$, the measured stress will not fall to zero instantaneously, but will decay in an exponential manner.



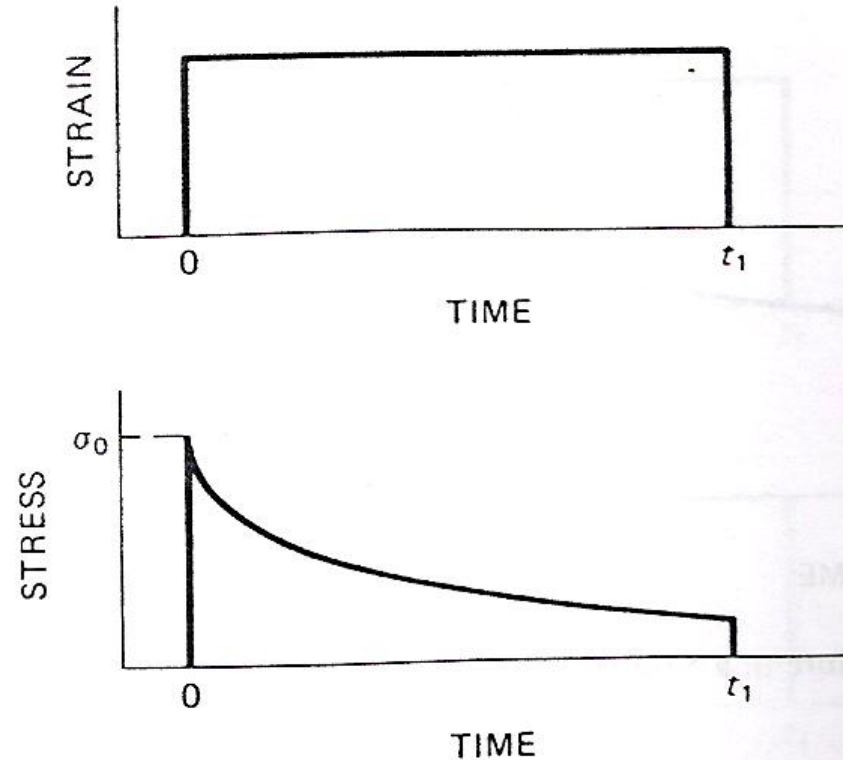
- Relaxation is slower for Polymer B than for Polymer A, as a result of greater elasticity.
- These differences may arise from polymer microstructure (molecular weight, branching).

CREEP



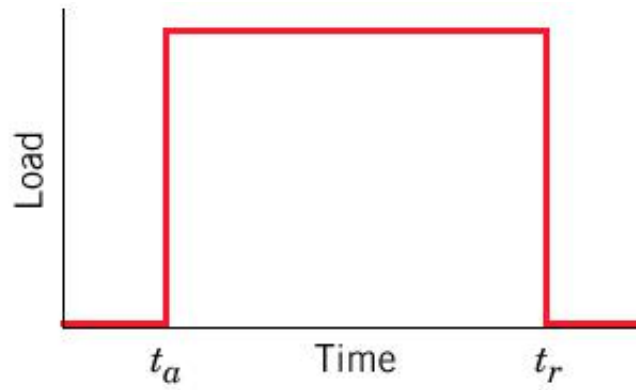
Constant stress is applied → the strain relaxes as function of time

STRESS RELAXATION

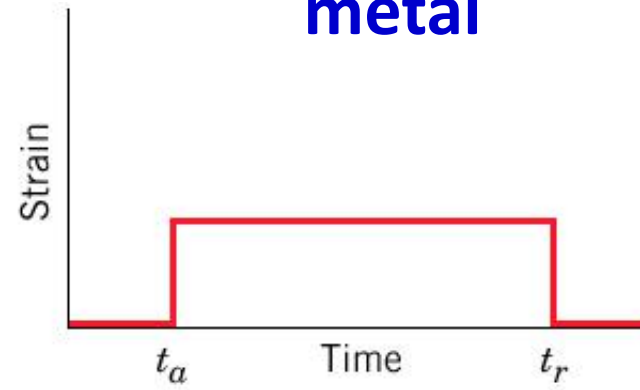


Constant strain is applied → the stress relaxes as function of time

metal

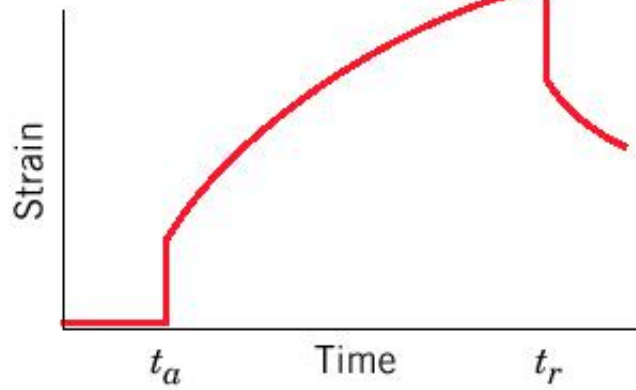


(a)



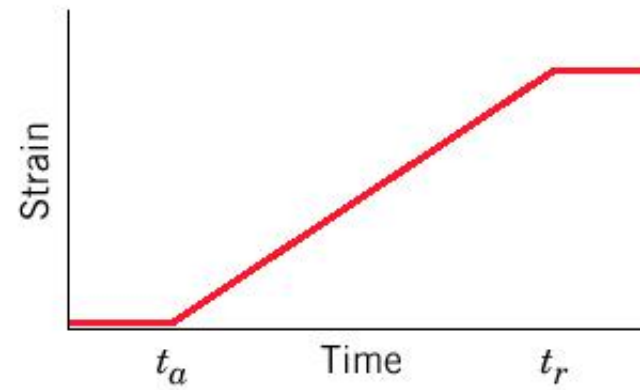
(b)

elastomer



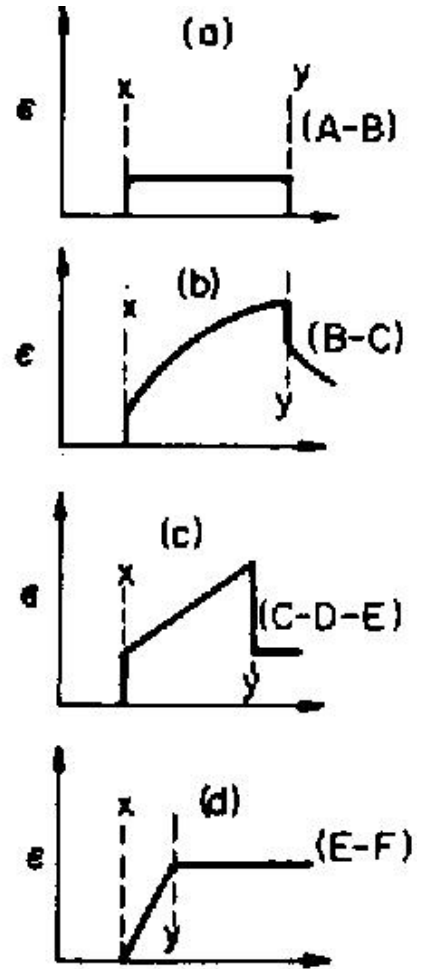
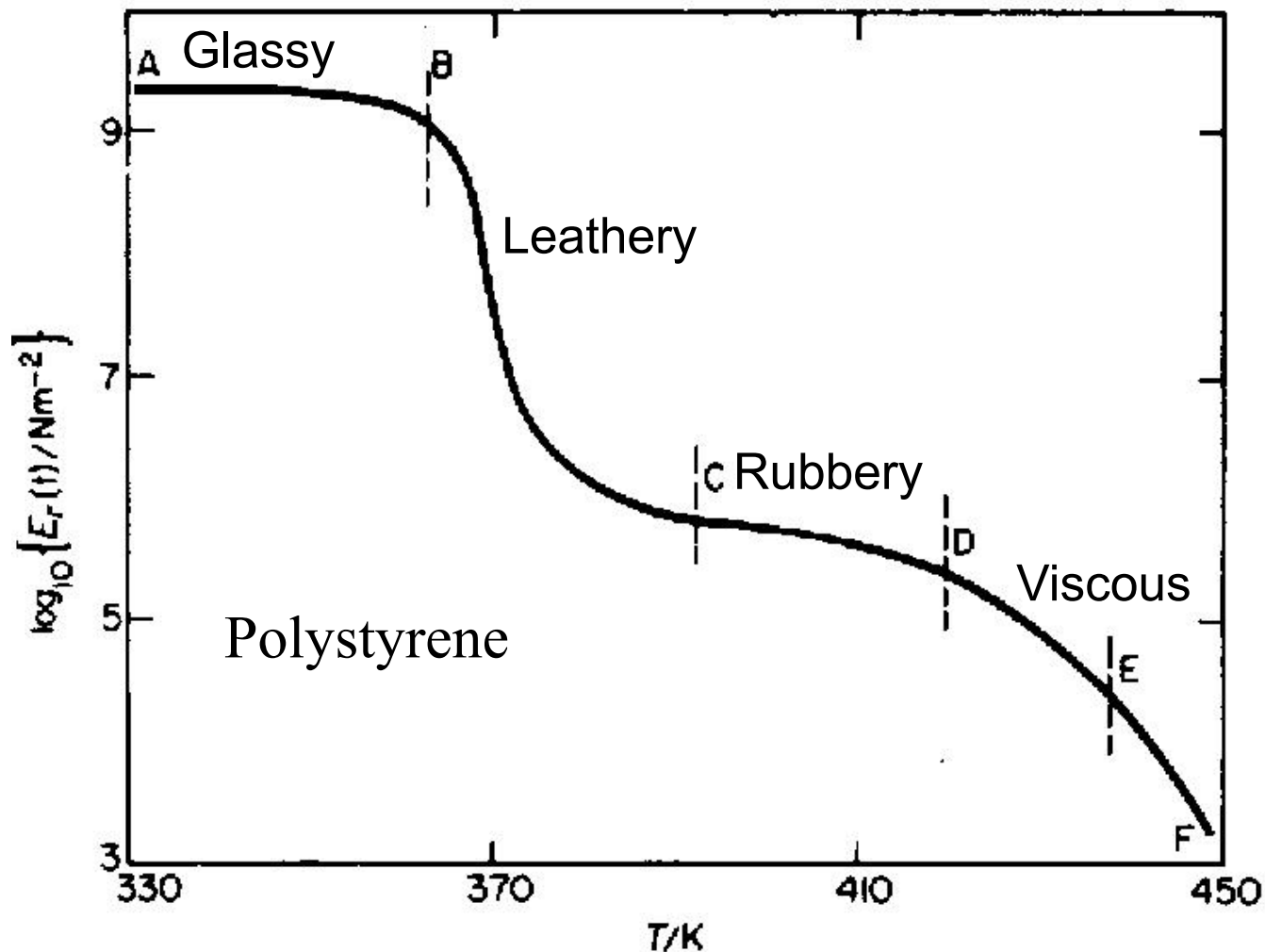
(c)

Viscous liquid



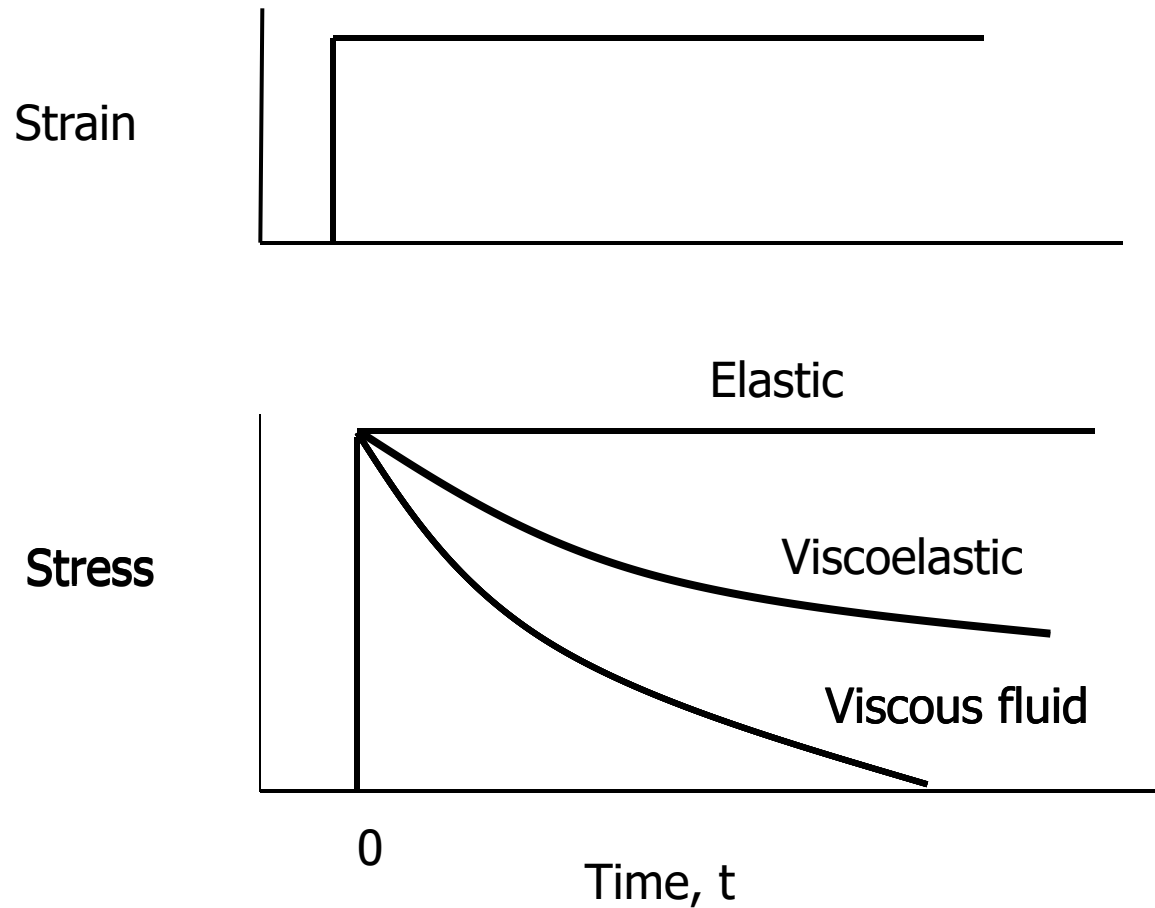
(d)

Static Modulus of Amorphous PS



Stress applied at x and removed at y

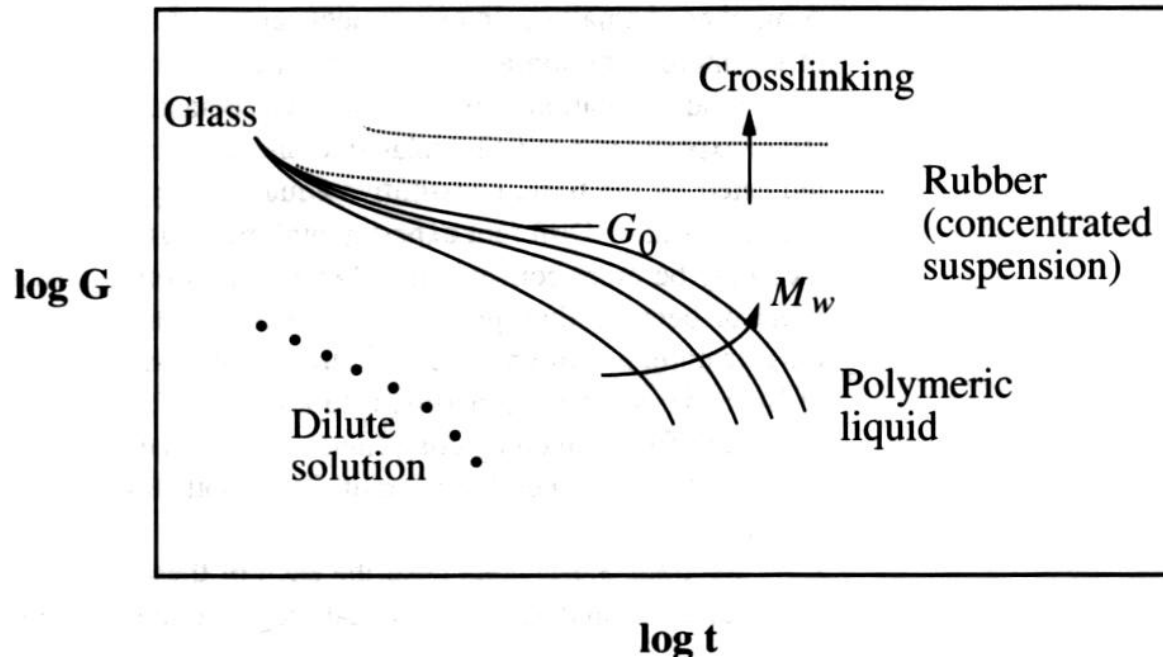
Stress Relaxation Test



Stress relaxation

Stress relaxation after a step strain γ_0 is the fundamental way in which we define the relaxation modulus:

$$G(t) = \frac{\tau(t)}{\gamma_0}$$



➤ G_0 (or G_N^0) is the “plateau modulus”:

$$G_N^0 = \frac{\rho RT}{M_e}$$

where M_e is the average mol. weight between entanglements

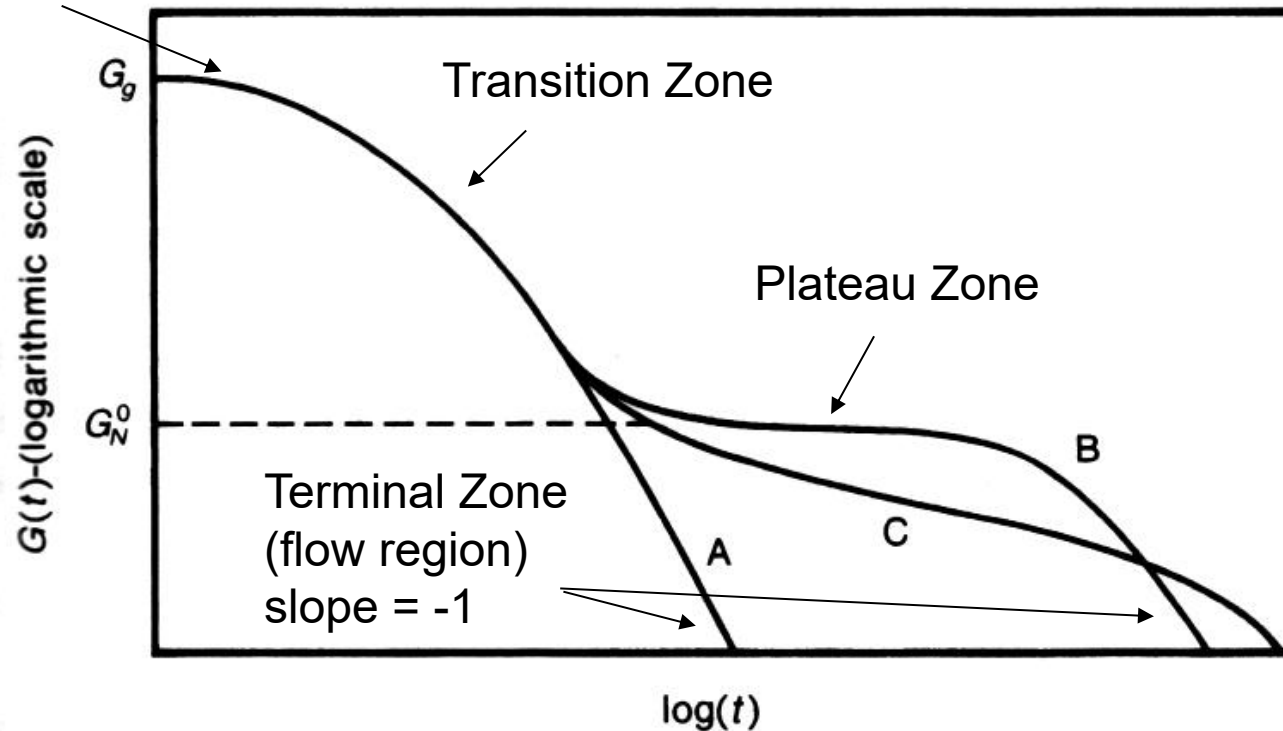
➤ $G(t)$ is defined for shear flow. We can also define a relaxation modulus for extension:

$$E(t) = \frac{\sigma(t)}{\epsilon_0}$$

τ stress γ strain η viscosity G modulus

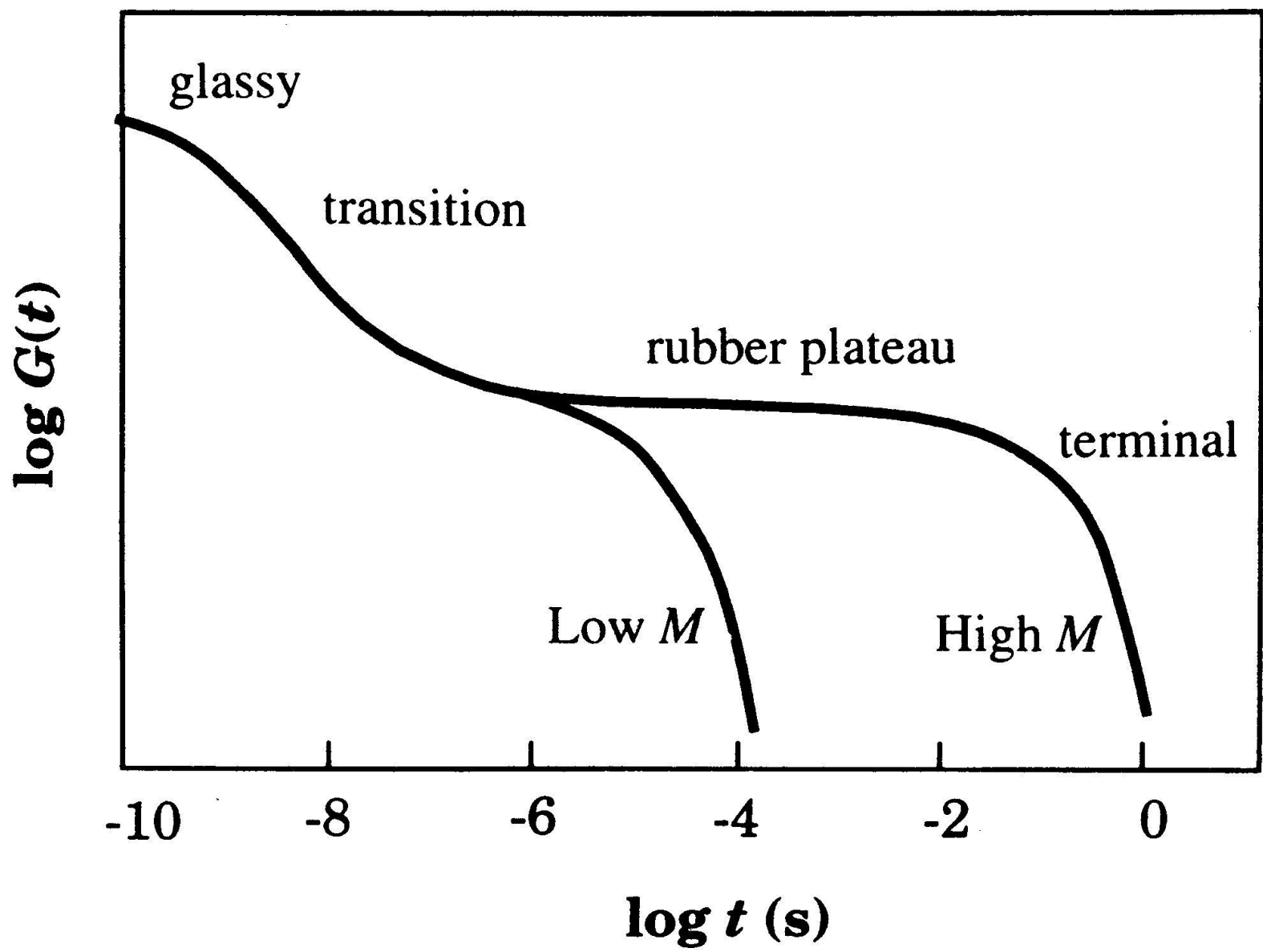
Stress relaxation of an uncrosslinked melt

Glassy behavior



2-3. Relaxation modulus for three samples of a typical linear polymer. A is monodisperse with $M < M_C$; B is monodisperse with $M \gg M_C$, and C is polydisperse with $M_w \gg M_C$.

M_C : critical molecular weight above which entanglements exist



Mechanical Model

- Methods that used to predict the behaviour of visco-elasticity.
- They consist of a combination of between elastic behaviour and viscous behaviour.
- **Two basic elements that been used in this model:**
 1. **Elastic spring** with modulus which follows Hooke's law
 2. **Viscous dashpots** with viscosity η which follows Newton's law.
- The models are used to explain the phenomena creep and stress relaxation of polymers involved with different combination of this two basic elements.

Dynamic Viscosity (dashpot)

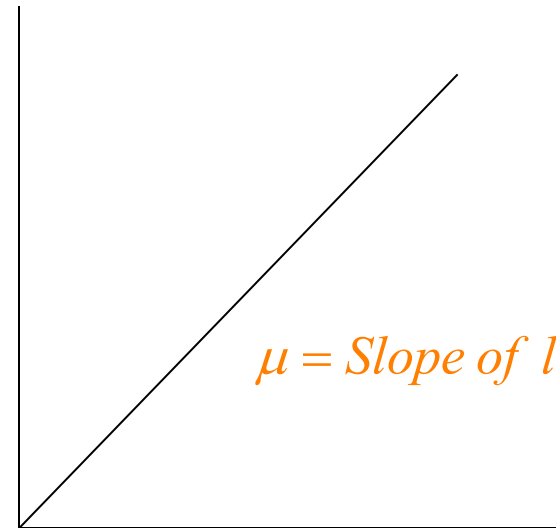
- Lack of slipperiness
- Resistance to flow
- Interlayer friction

$$\mu = \frac{\tau}{\dot{\gamma}}$$

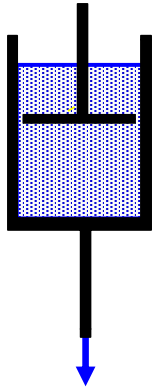
SI Unit: Pascal-second

1 centi-Poise = milli Pascal-second

Shear stress



Shear rate



τ

stress

γ

strain

η

viscosity

G

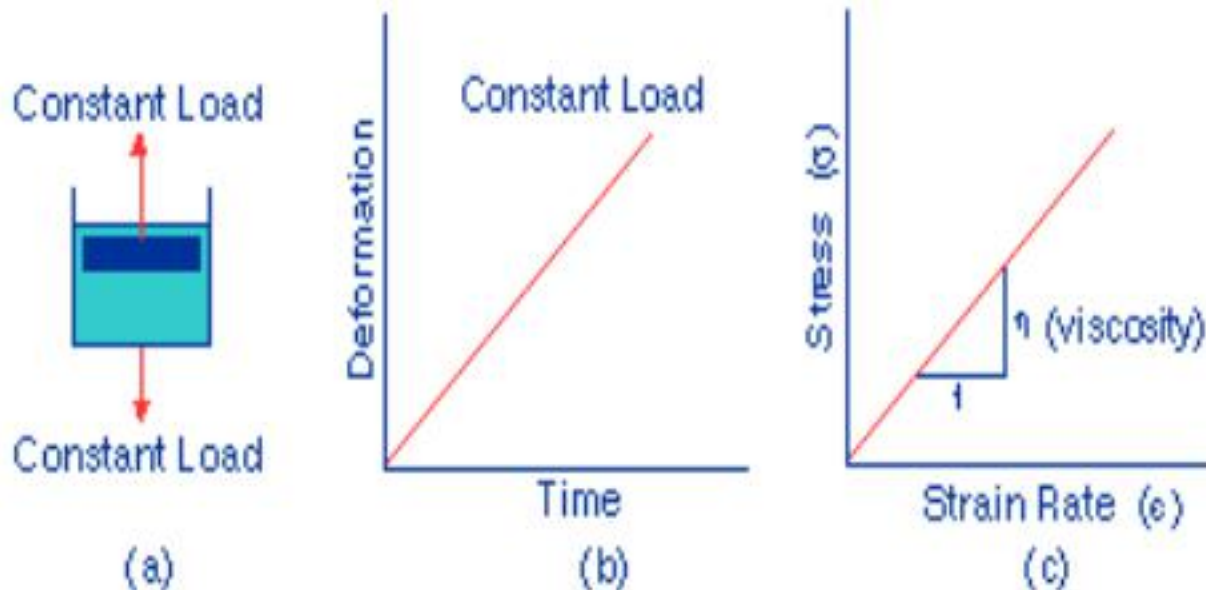
modulus

Ideal Liquid

η = viscosity

de/dt = strain rate

$$\sigma = \eta \frac{de}{dt}$$



The viscous response is generally time- and rate-dependent.

Ideal Liquid

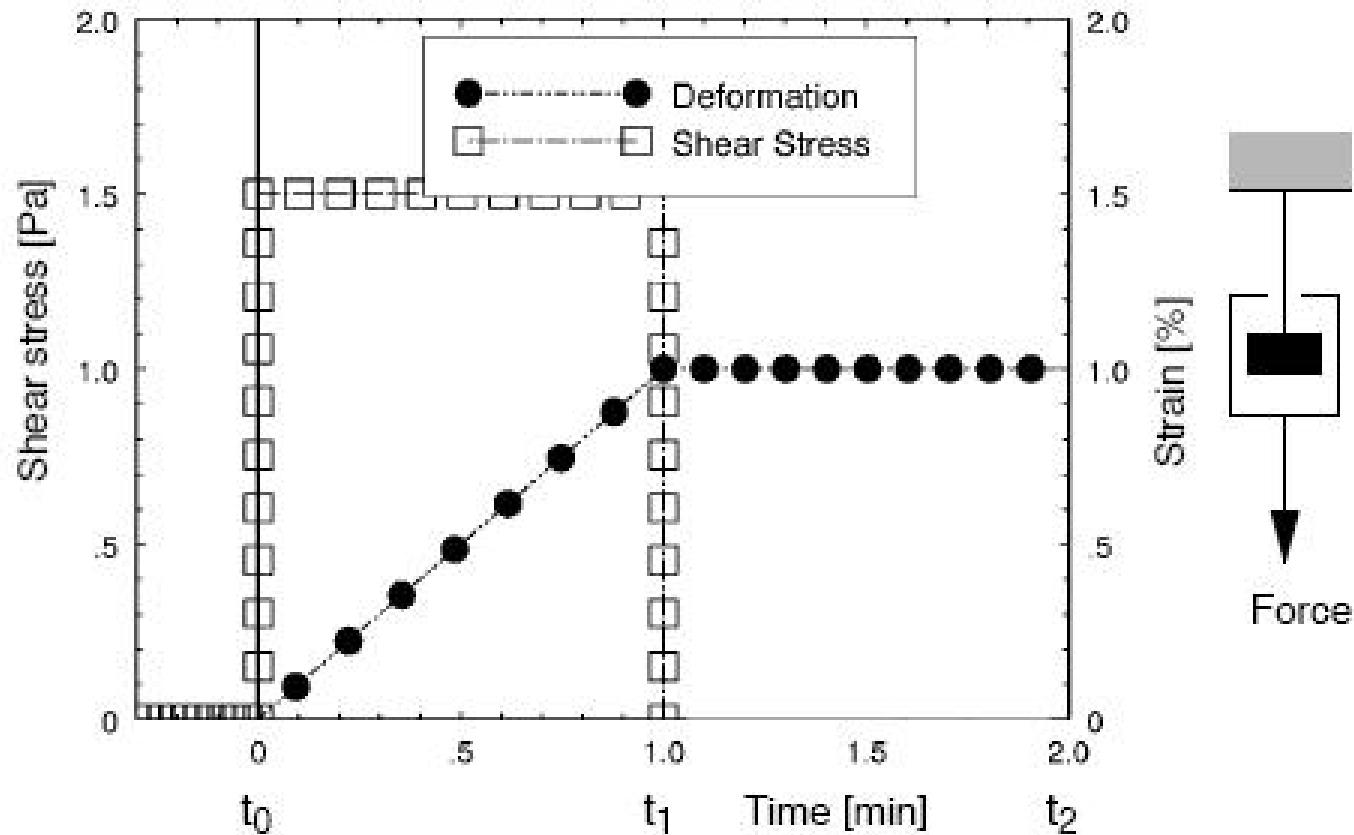
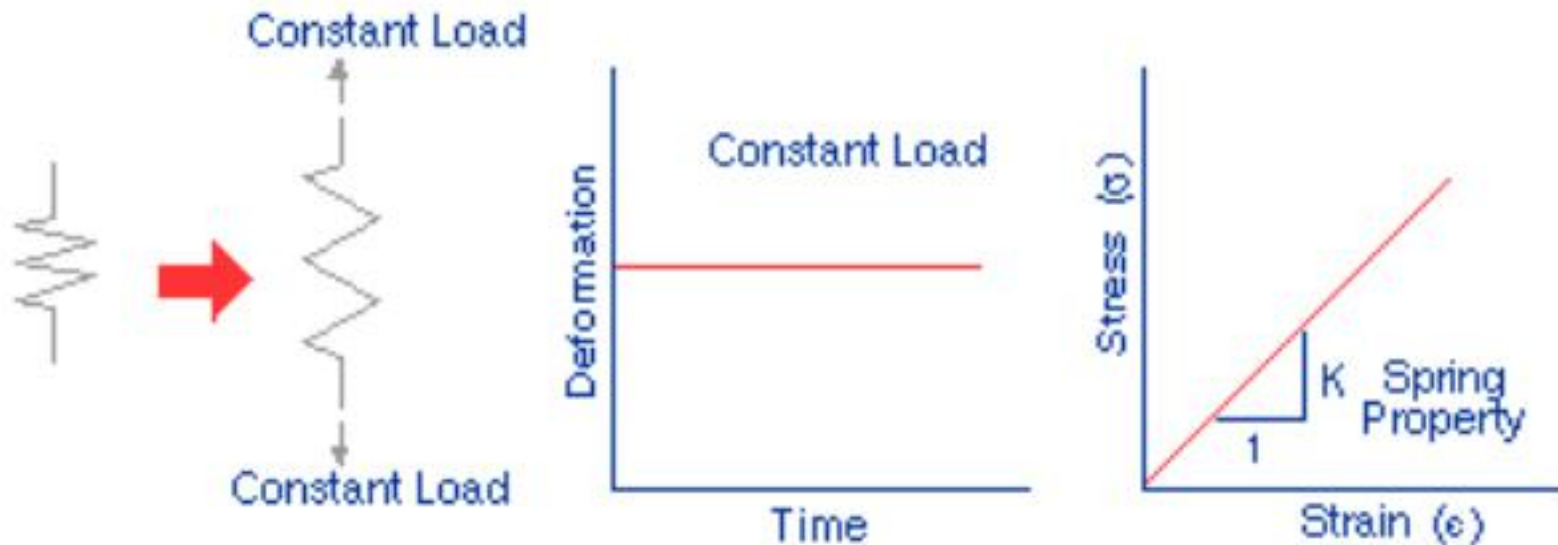


Fig. 59 Shear stress and deformation of an ideal liquid versus time.

Ideal (elastic) Solid

$$\sigma = E\varepsilon \quad \text{Hooks Law}$$



response is independent of time and the deformation is dependent on the spring constant.

Ideal Solid

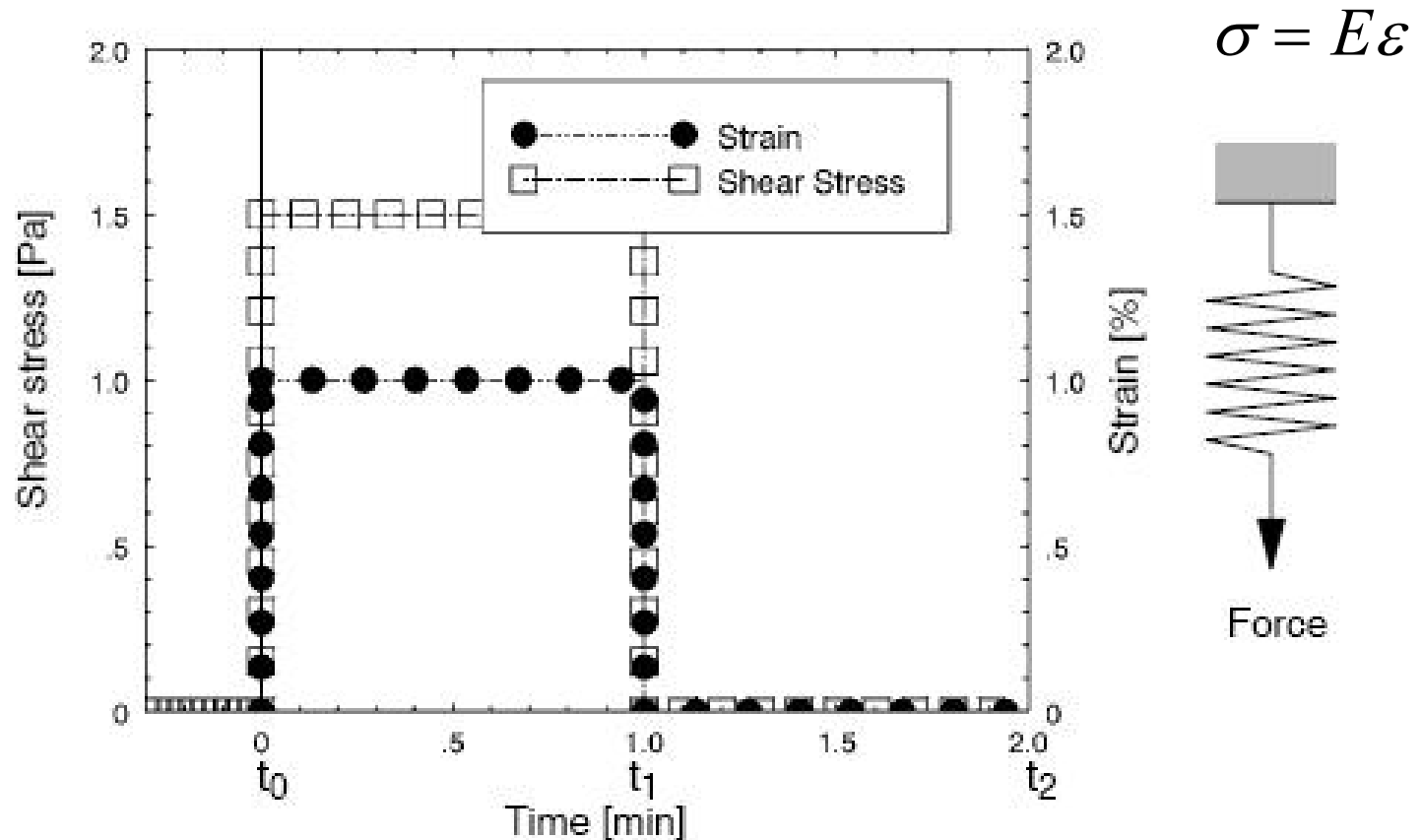
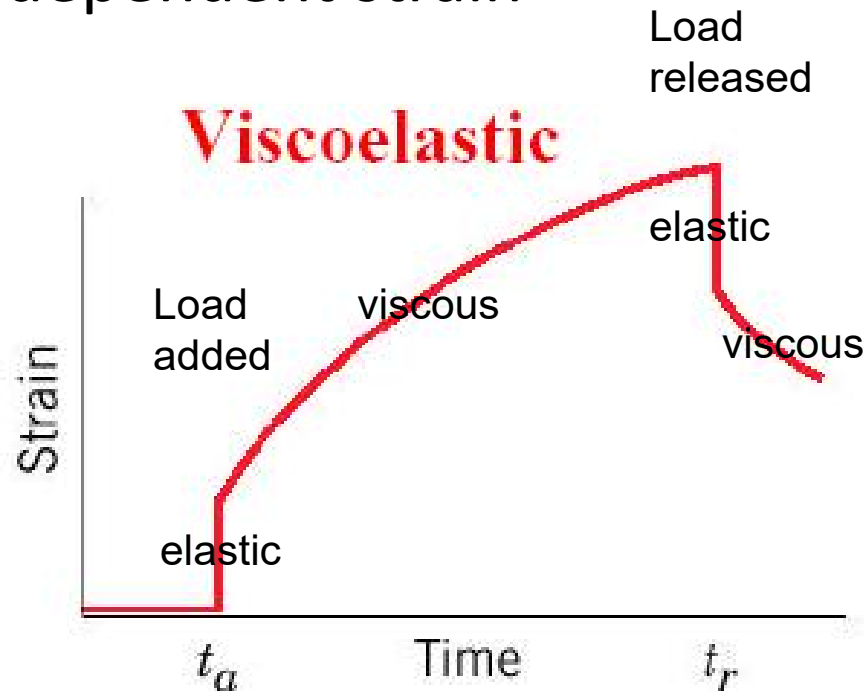
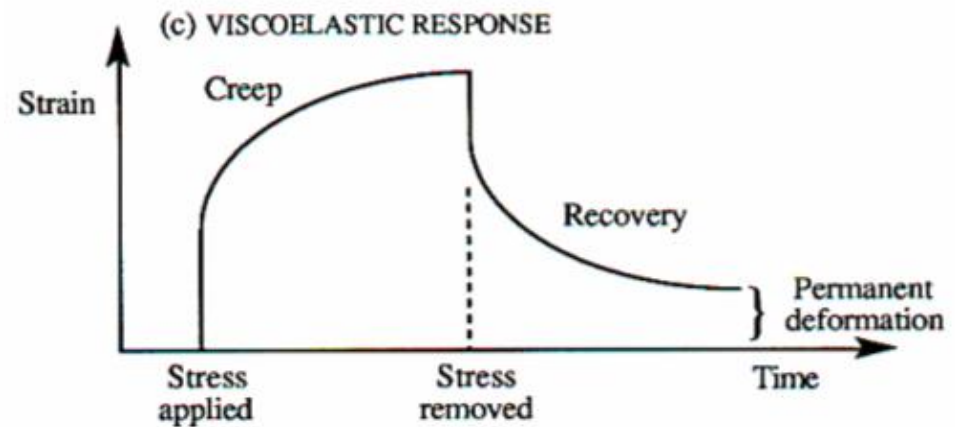
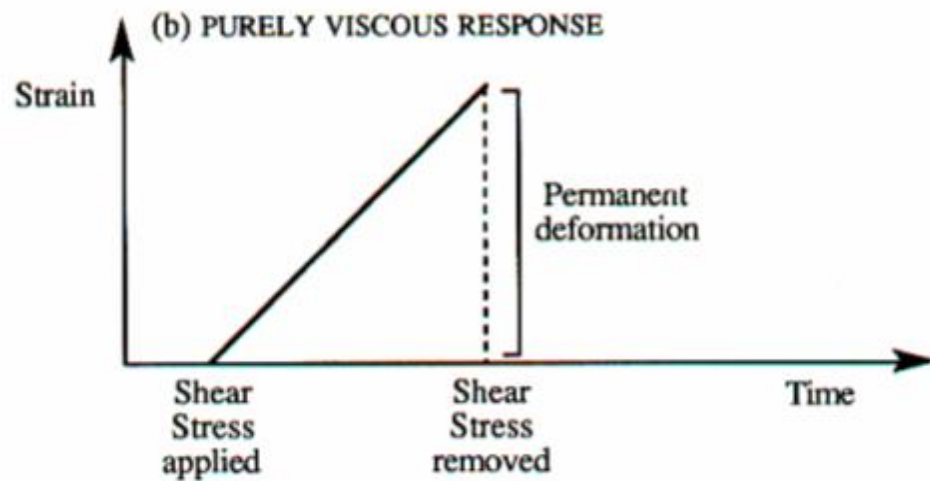
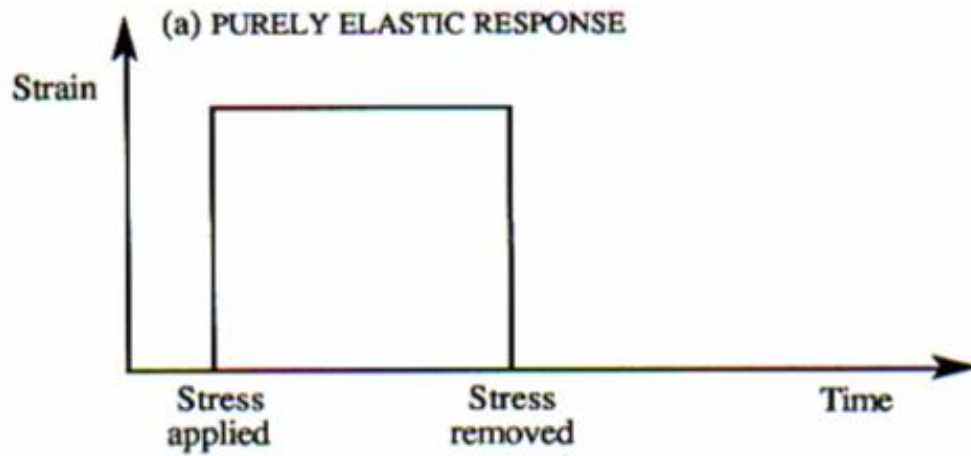


Fig. 58 Shear stress and strain of an ideal solid versus time.

Polymer is called visco- elastic because:

- Showing both behaviour elastic & viscous behaviour
- Instantaneously elastic strain followed by viscous time dependent strain





τ stress γ strain η viscosity G modulus

Maxwell Model

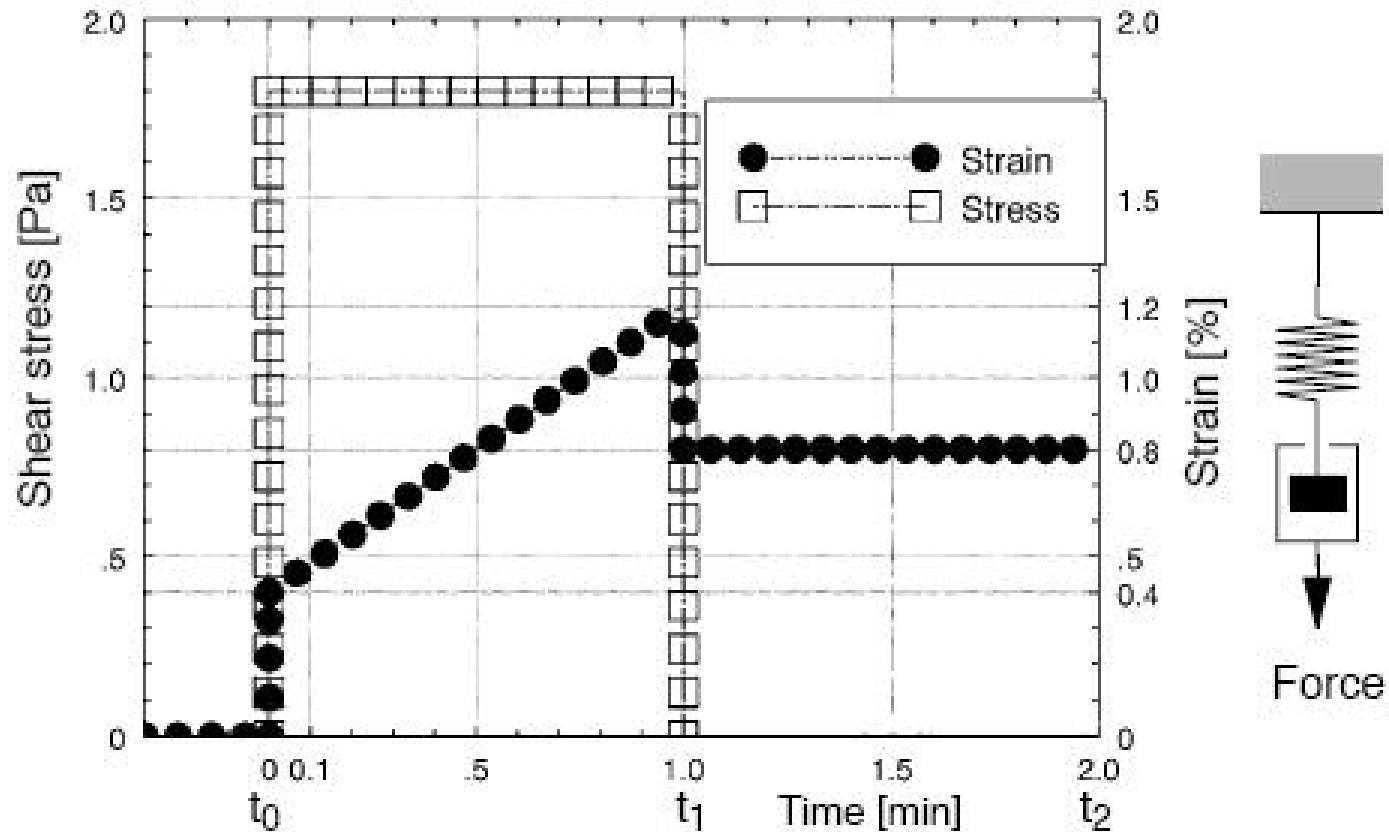


Fig. 61: Creep and recovery of a Maxwell liquid versus

Kelvin Voigt Model

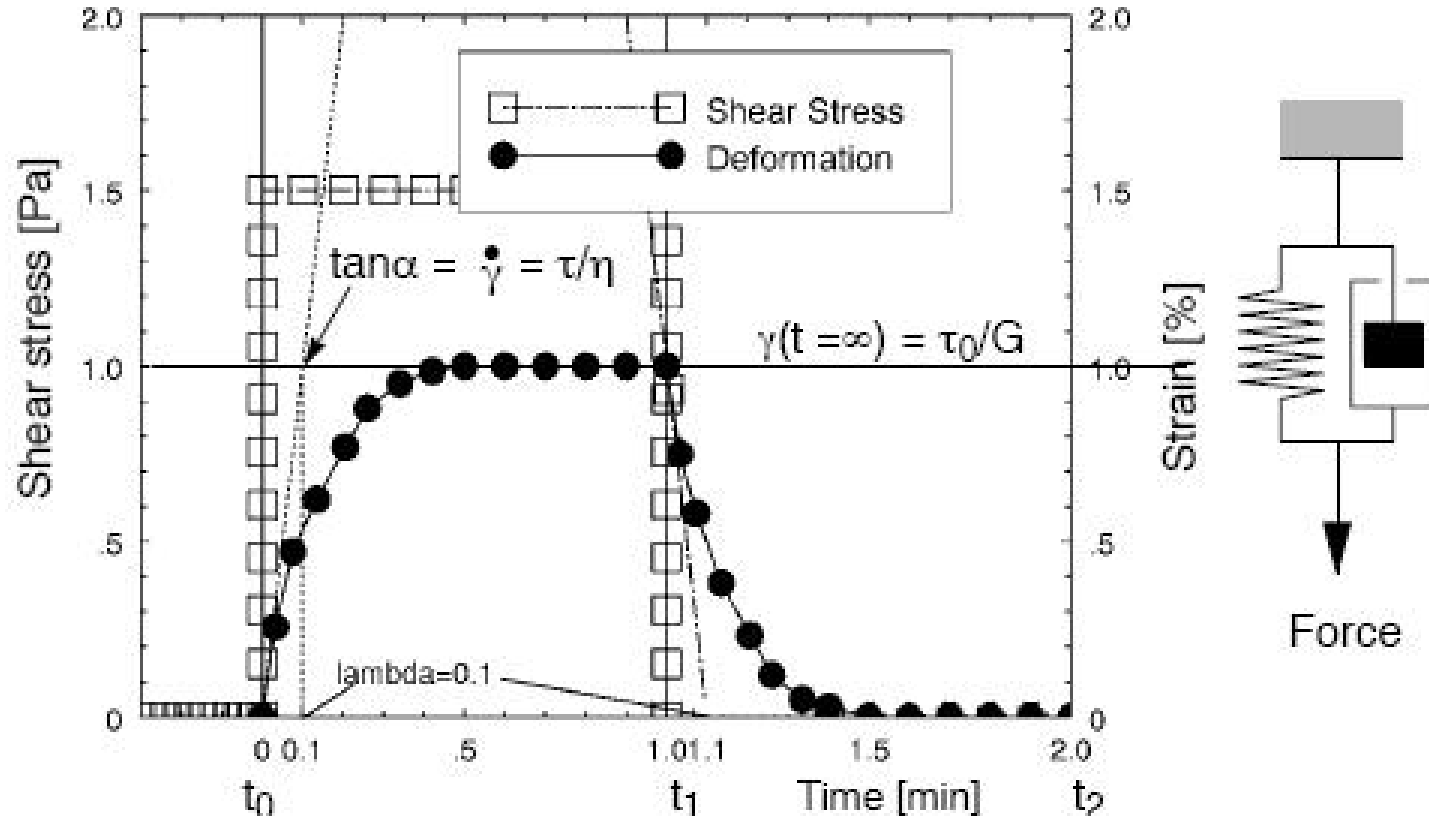


Fig. 60 Creep and recovery of a Voigt solid versus time

Burger Model

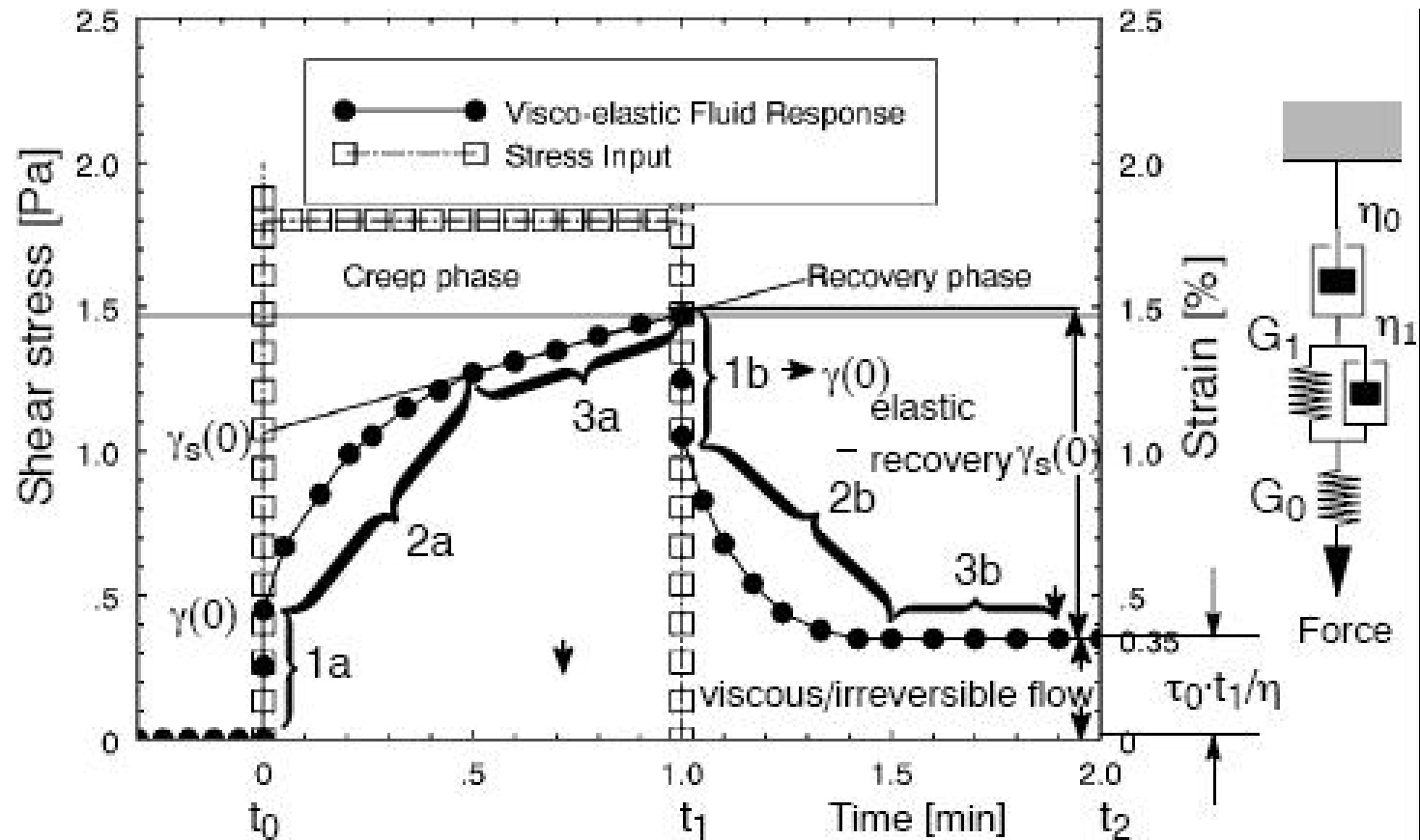
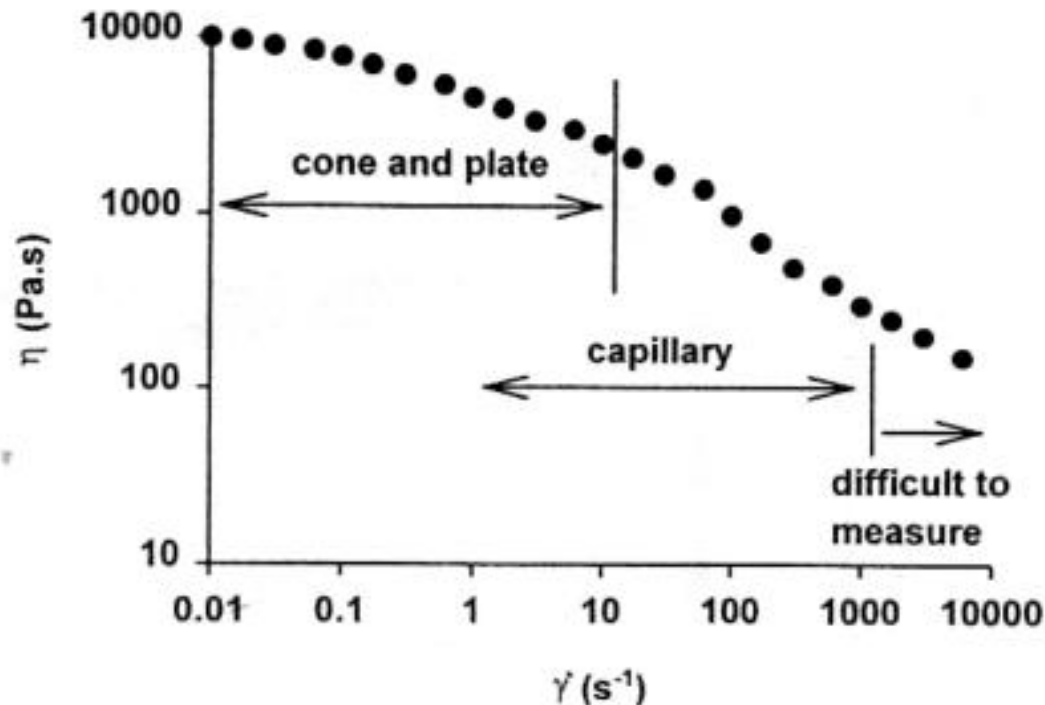


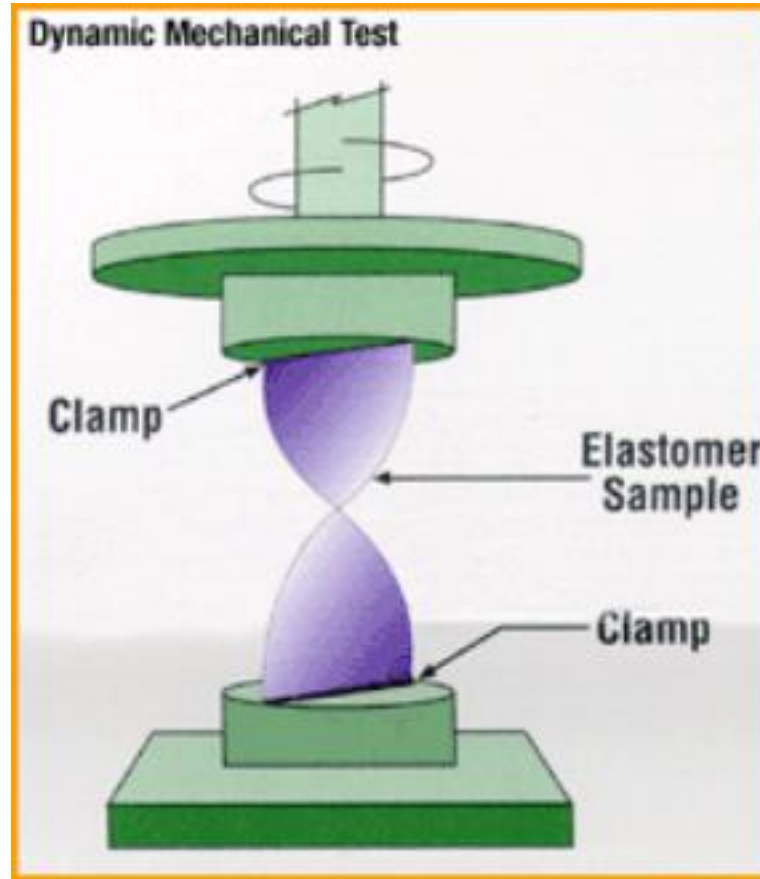
Fig. 62: Creep and recovery of a Burger model as a function of time

Measurements of Shear Viscosity

- Melt Flow Index
- Capillary Rheometer
- Coaxial Cylinder Viscometer (Couette)
- Cone and Plate Viscometer (Weissenberg rheogoniometer)
- Disk-Plate (or parallel plate) viscometer



Dynamic Mechanical Analysis



Spring Model

$$\gamma = \gamma_0 \cdot \sin(\omega \cdot t)$$

γ_0 = maximum strain

ω = angular velocity

Since stress, τ , is

$$\tau = G\gamma$$

$$\tau = G\gamma_0 \sin(\omega t)$$

And τ and γ
are in
phase

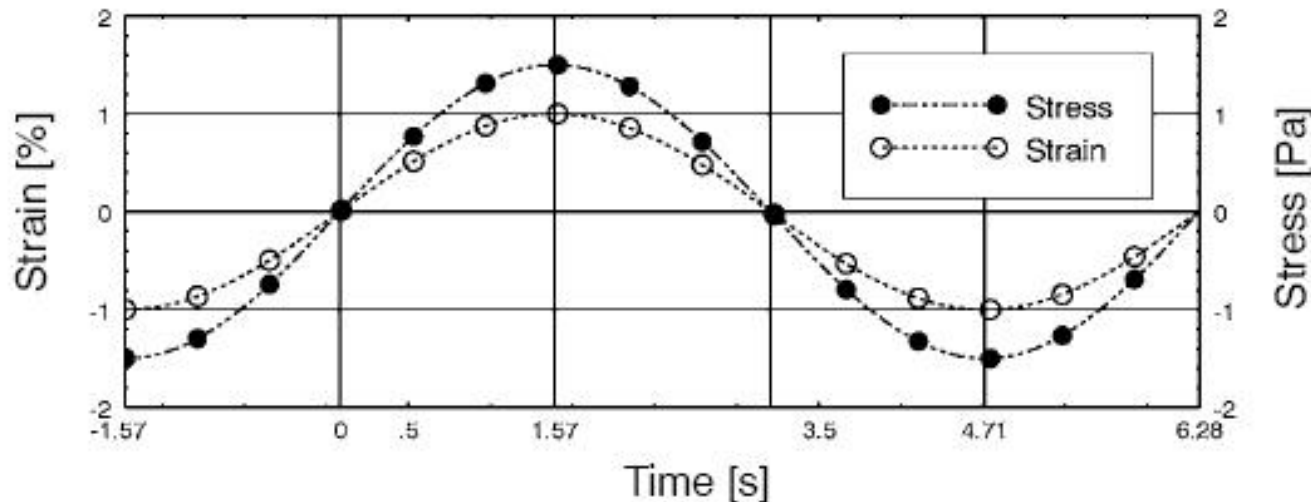
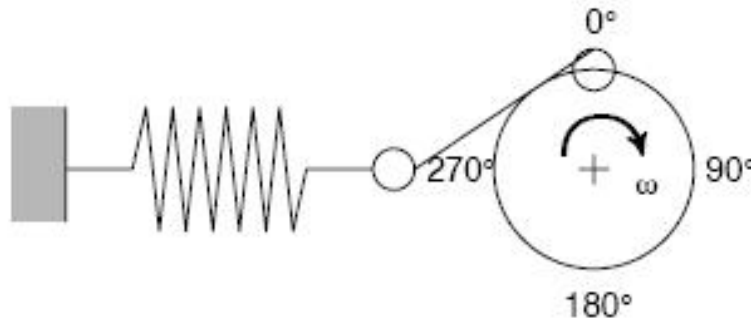


Fig.66 Dynamic test: Strain input and stress response of a spring-type solid.

Dashpot Model

$$\tau_{\text{dashpot}} = \eta \dot{\gamma} = \eta \gamma_0 \omega \cos(\omega t)$$

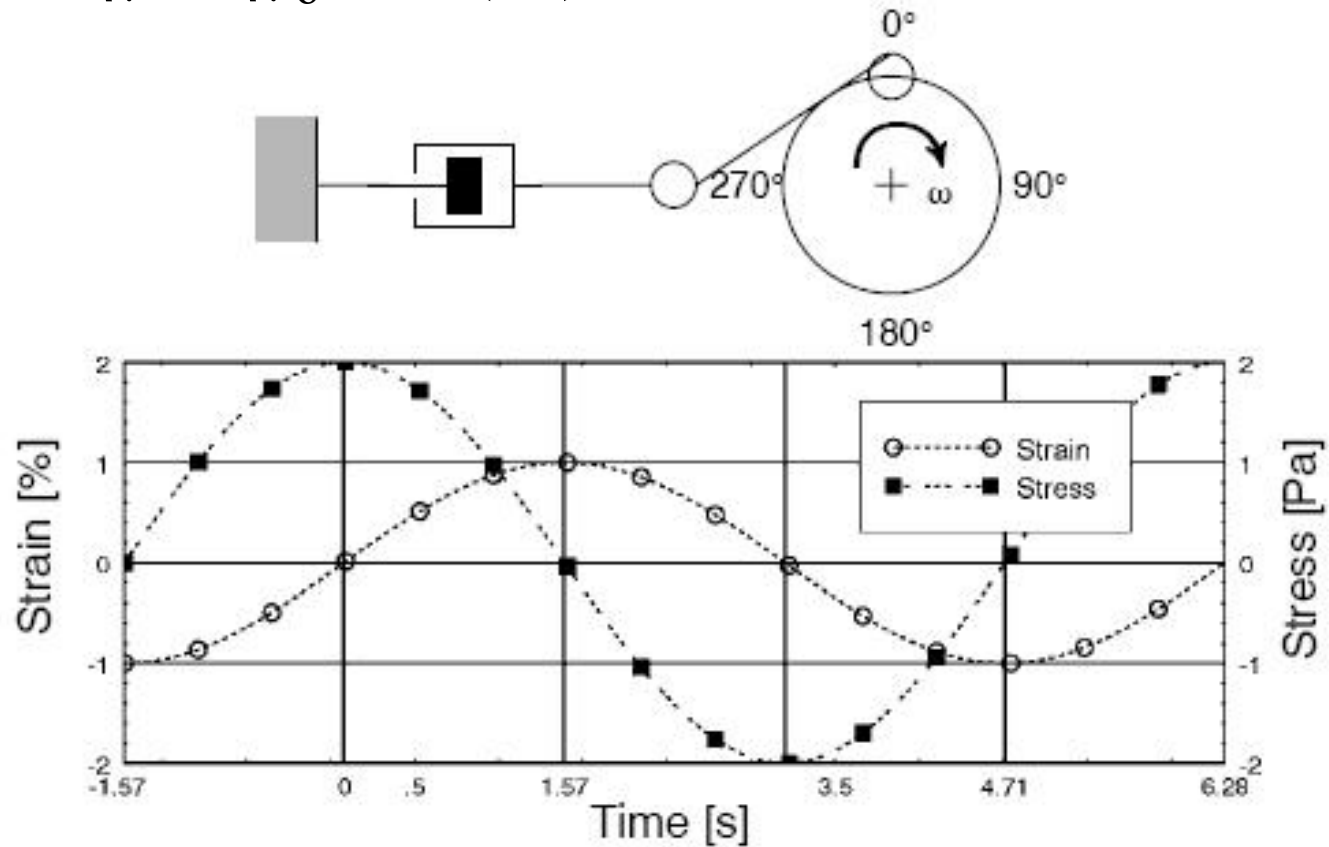


Fig.67 Dynamic test: Strain input and stress response of a Newtonian liquid or a dashpot

Kelvin-Voigt Model

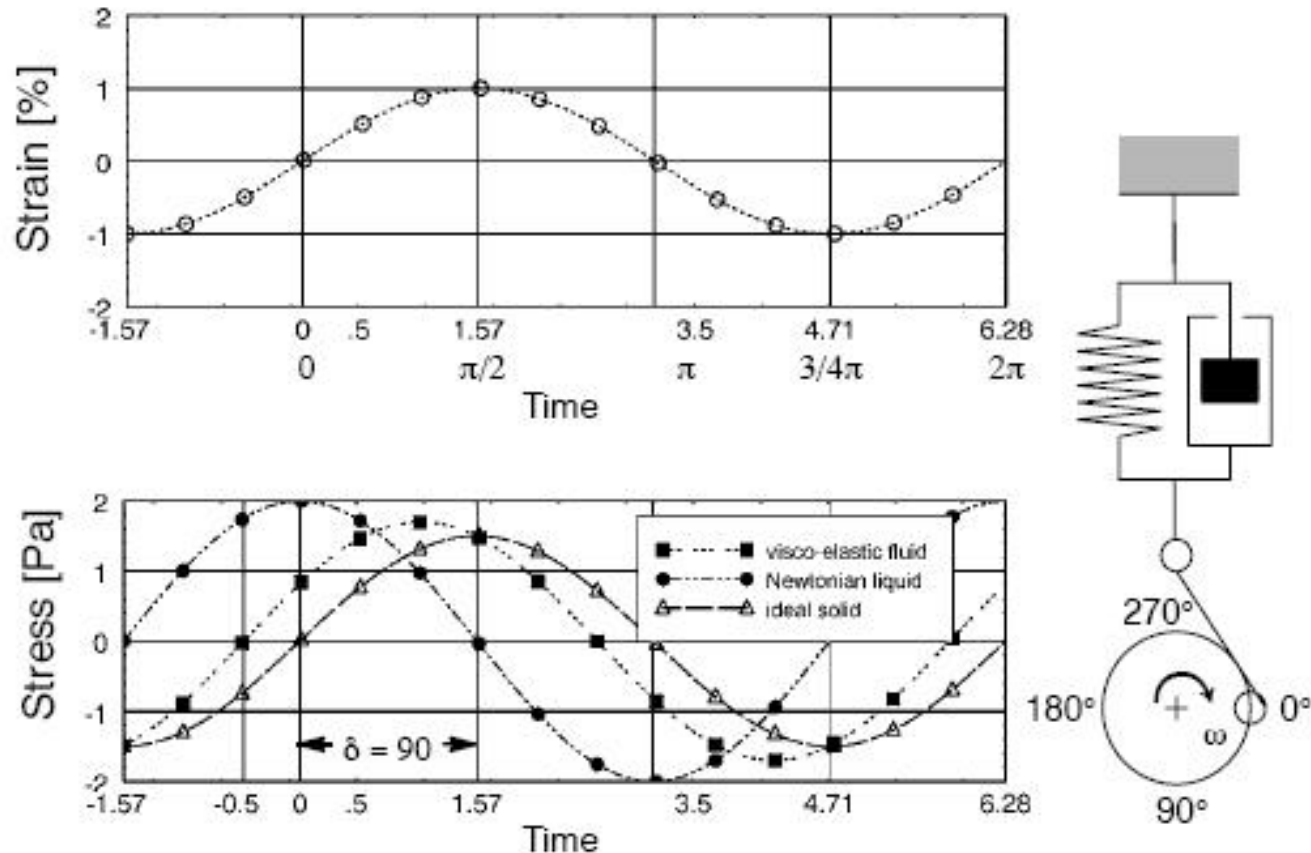
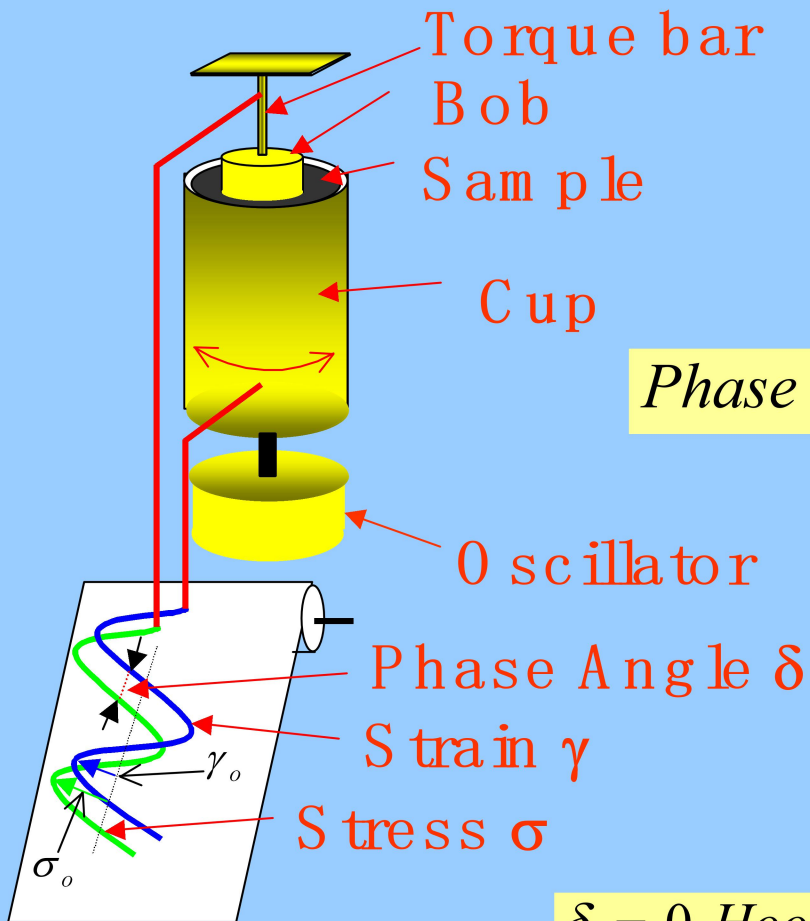


Fig.68 CR-dynamic tests of a visco-elastic fluid, a Newtonian liquid and an ideal solid

Viscoelastic Measurements



Phase Angle = δ

Storage Modulus

$$G'(\omega) = \frac{\sigma_0 \cos \delta}{\gamma_0}$$

Loss Modulus

$$G''(\omega) = \frac{\sigma_0 \sin \delta}{\gamma_0}$$

Loss Tangent

$$\tan \delta = \frac{G''(\omega)}{G'(\omega)}$$

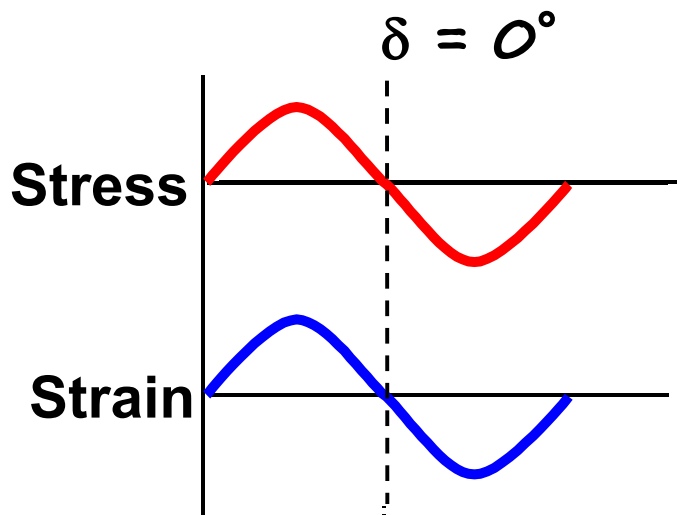
$\delta = 0$ Hookean Elastic Solid

$0 < \delta < 90^\circ$ Viscoelastic Material

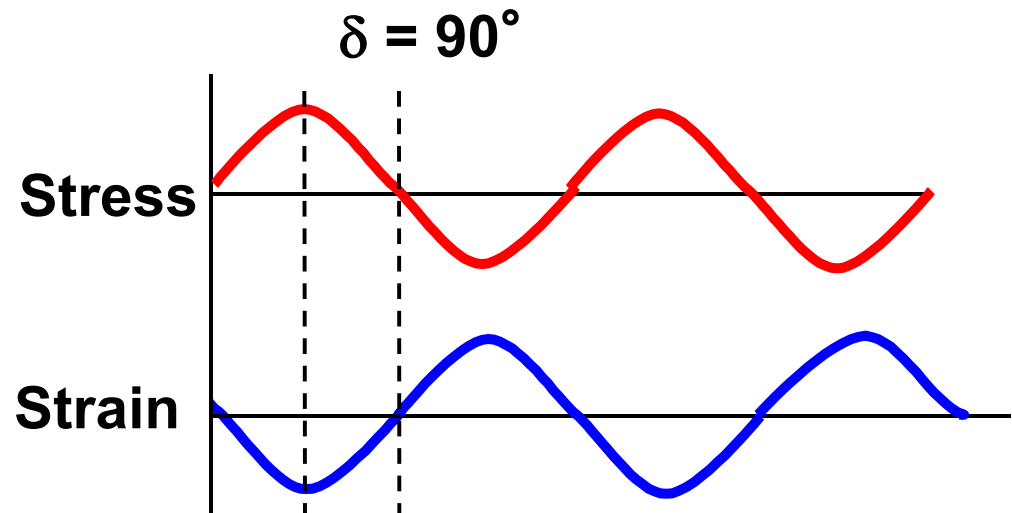
$\delta = 90^\circ$ Viscous Liquid

Response for Classical Extremes

Purely Elastic Response (Hookean Solid)

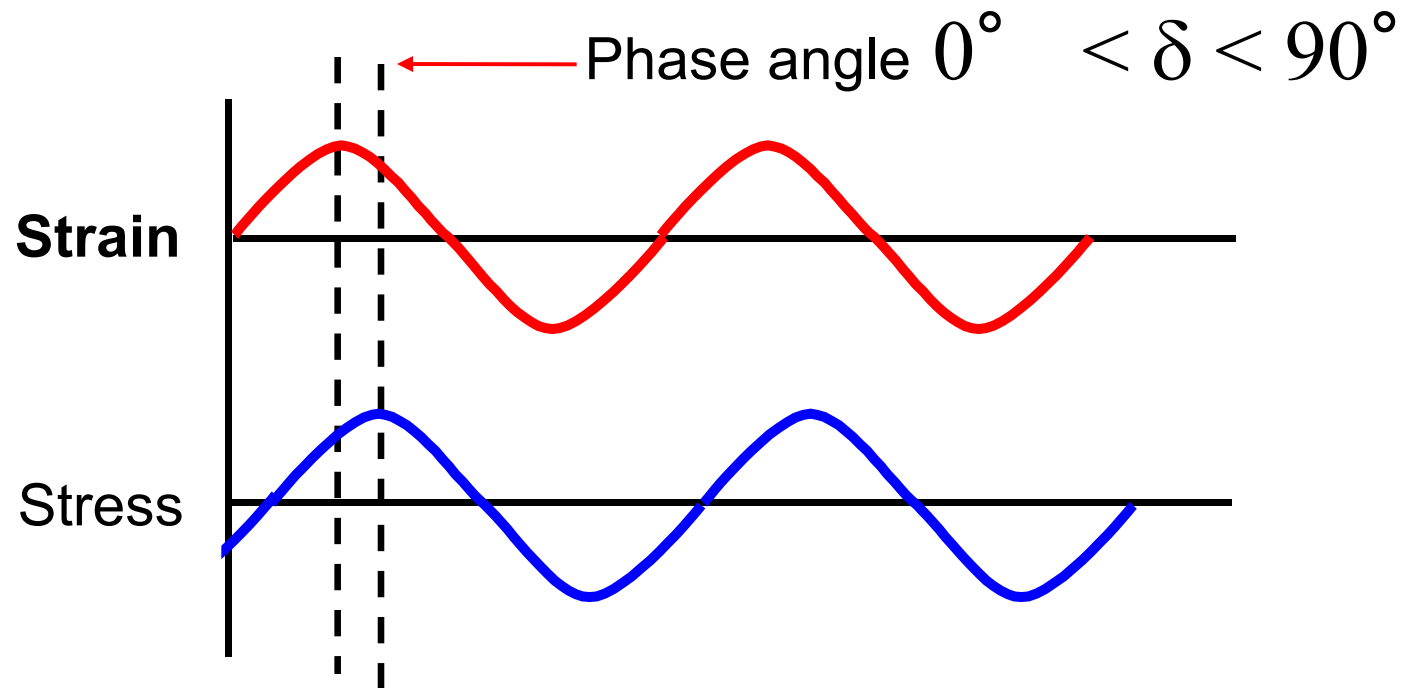


Purely Viscous Response (Newtonian Liquid)



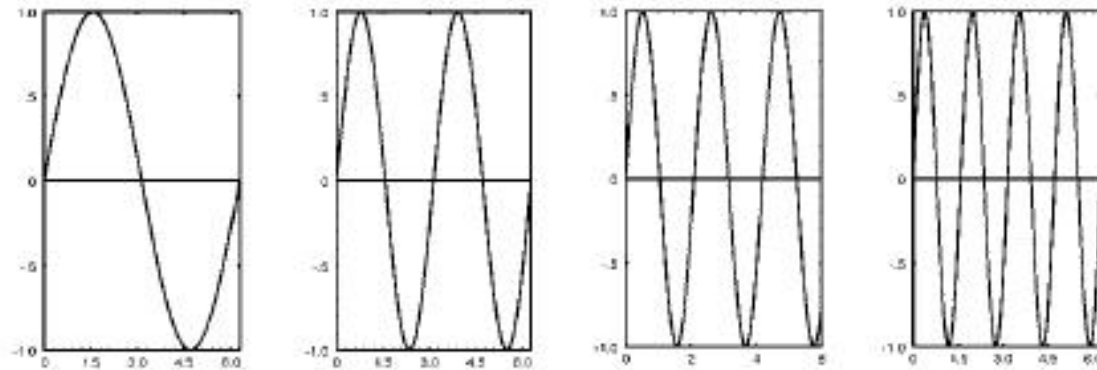
Courtesy: TA Instruments

Viscoelastic Material Response



Courtesy: TA Instruments

Real Visco-Elastic Samples



Frequencies are stepwise increased: Each step provides pairs of G^* and δ

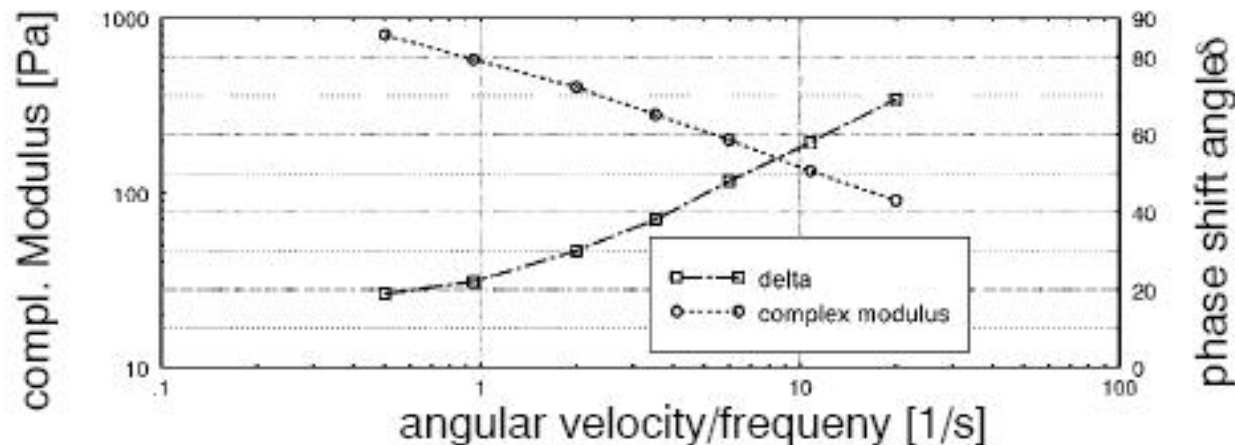
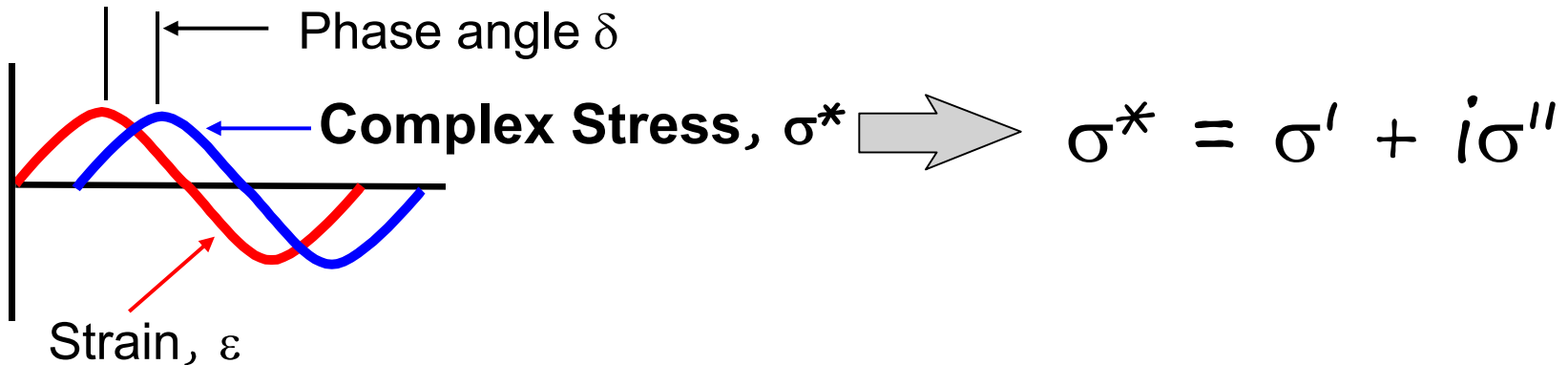


Fig. 71 Dynamic test: Autom. sweep of the angular velocity/frequency

DMA Viscoelastic Parameters:

The Complex, Elastic, & Viscous Stress

- The stress in a dynamic experiment is referred to as the complex stress σ^*
- The complex stress can be separated into two components:
 - 1) An elastic stress in phase with the strain. $\sigma' = \sigma^* \cos \delta$
 σ' is the degree to which material behaves like an elastic solid.
 - 2) A viscous stress in phase with the strain rate. $\sigma'' = \sigma^* \sin \delta$
 σ'' is the degree to which material behaves like an ideal liquid.



DMA Viscoelastic Parameters

The Complex Modulus: Measure of materials overall resistance to deformation.

$$G^* = \text{Stress}^*/\text{Strain}$$

$$G^* = G' + iG''$$

The Elastic (Storage) Modulus:
Measure of elasticity of material. The ability of the material to store energy.

$$G' = (\text{stress}^*/\text{strain})\cos\delta$$

The Viscous (loss) Modulus:
The ability of the material to dissipate energy. Energy lost as heat.

$$G'' = (\text{stress}^*/\text{strain})\sin\delta$$

Tan Delta:
Measure of material damping - such as vibration or sound damping.

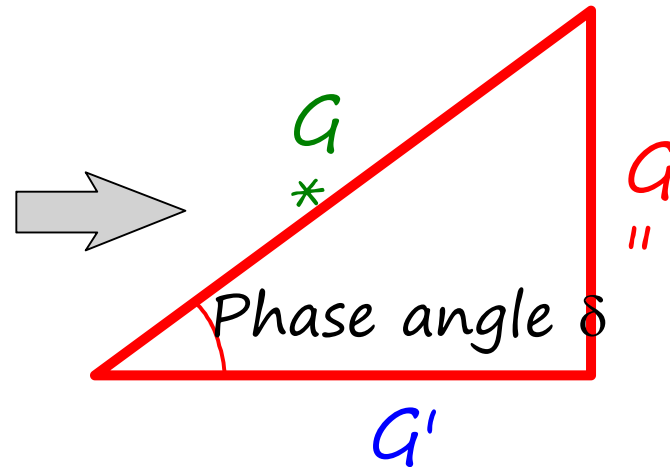
$$\tan \delta = G''/G'$$

DMA Viscoelastic Parameters: Damping, tan

Dynamic measurement represented as a vector

It can be seen here that

$$G^* = (G'^2 + G''^2)^{1/2}$$

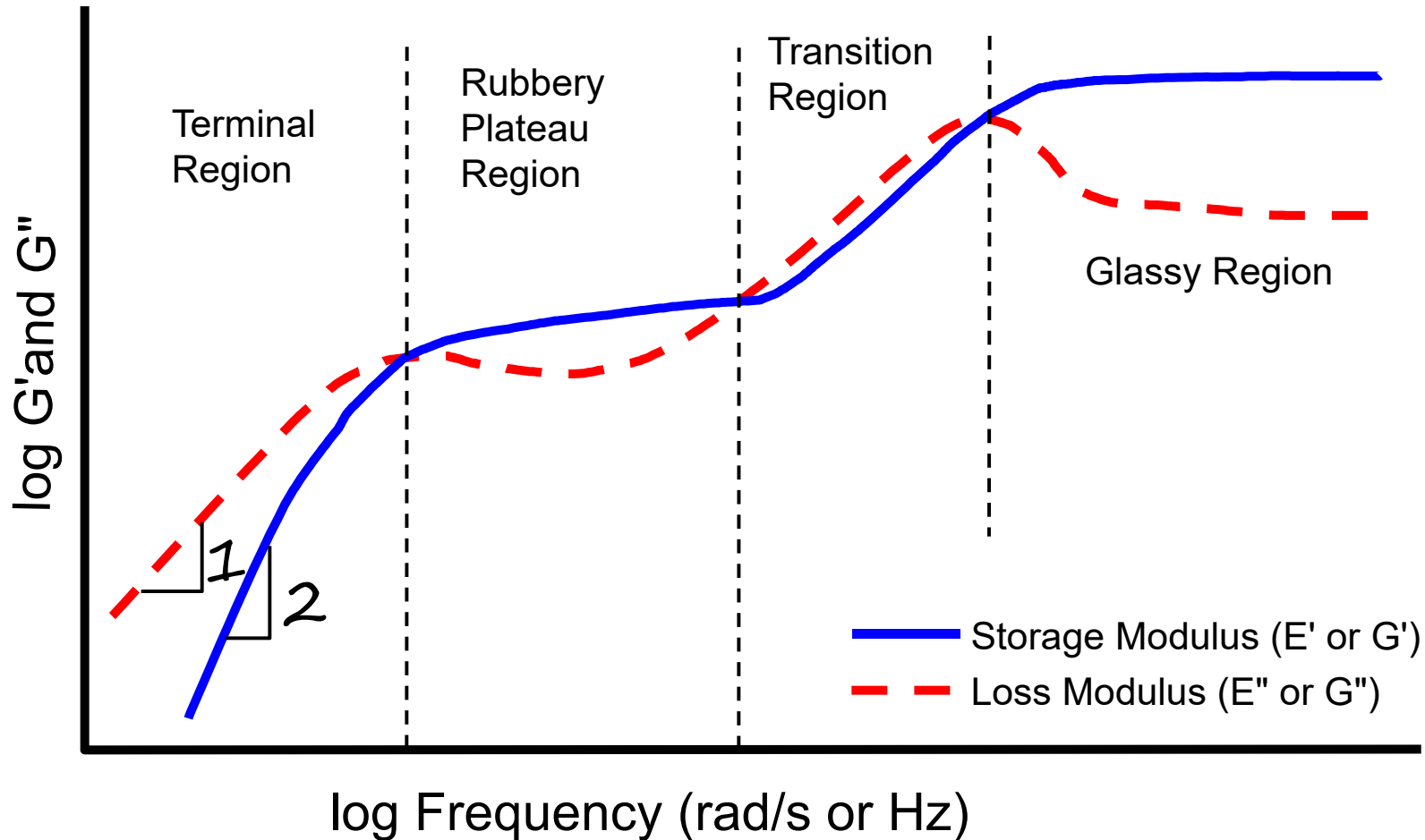


- The tangent of the phase angle is the ratio of the loss modulus to the storage modulus.

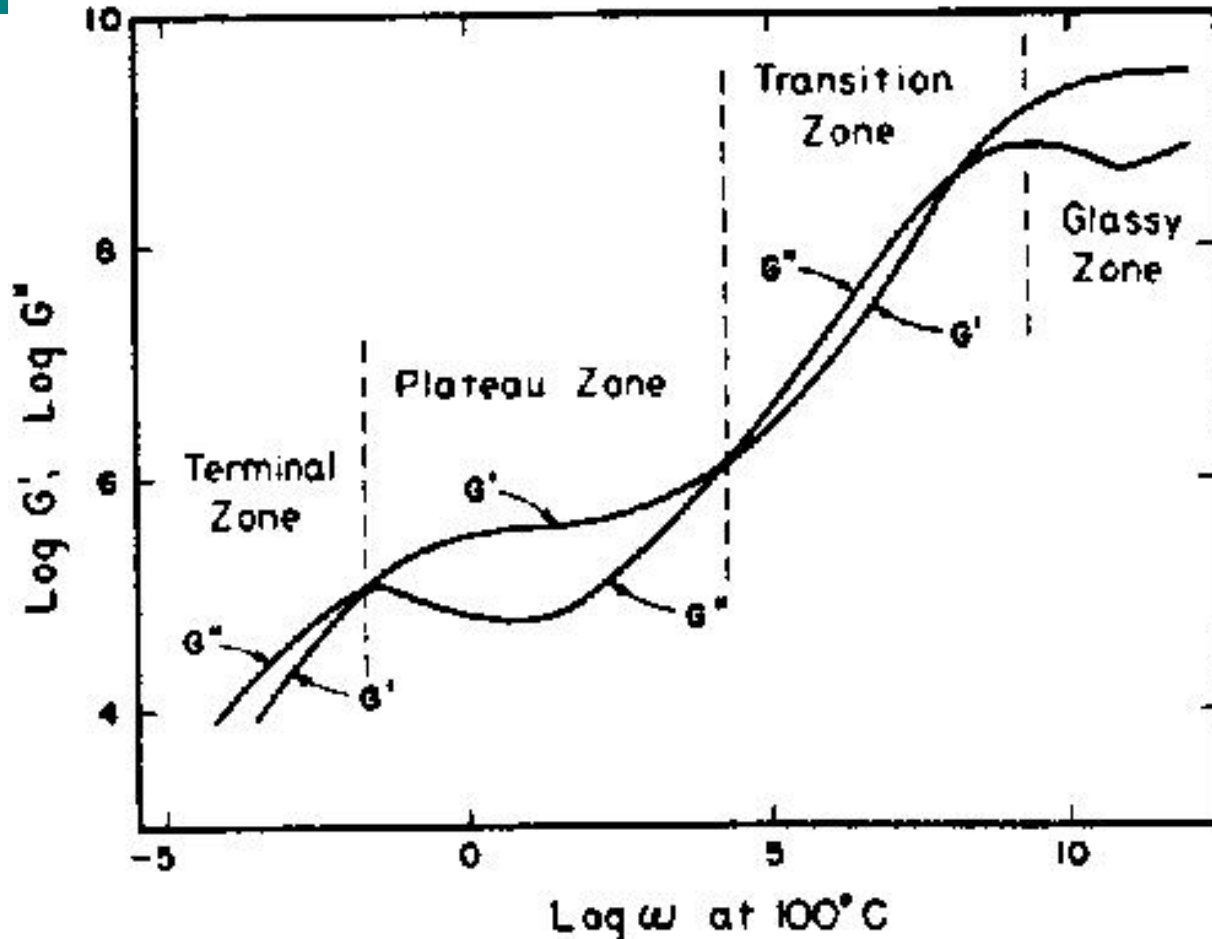
$$\tan \delta = G''/G'$$

- "TAN DELTA" ($\tan \delta$) is a measure of the damping ability of the material.

Frequency Sweep: Material Response



Viscoelasticity in Uncrosslinked, Amorphous Polymers



Logarithmic plots of G' and G'' against angular frequency for uncrosslinked poly(n-octyl methacrylate) at 100°C (above T_g), molecular weight 3.6×10^6 .

Dynamic Characteristics of Rubber Compounds

•Why do E' and E'' vary with frequency and temperature?

- The extent to which a polymer chains can store/dissipate energy depends on the rate at which the chain can alter its conformation and its entanglements relative to the frequency of the load.

•Terminal Zone:

- Period of oscillation is so long that chains can snake through their entanglement constraints and completely rearrange their conformations

•Plateau Zone:

- Strain is accommodated by entropic changes to polymer segments between entanglements, providing good elastic response

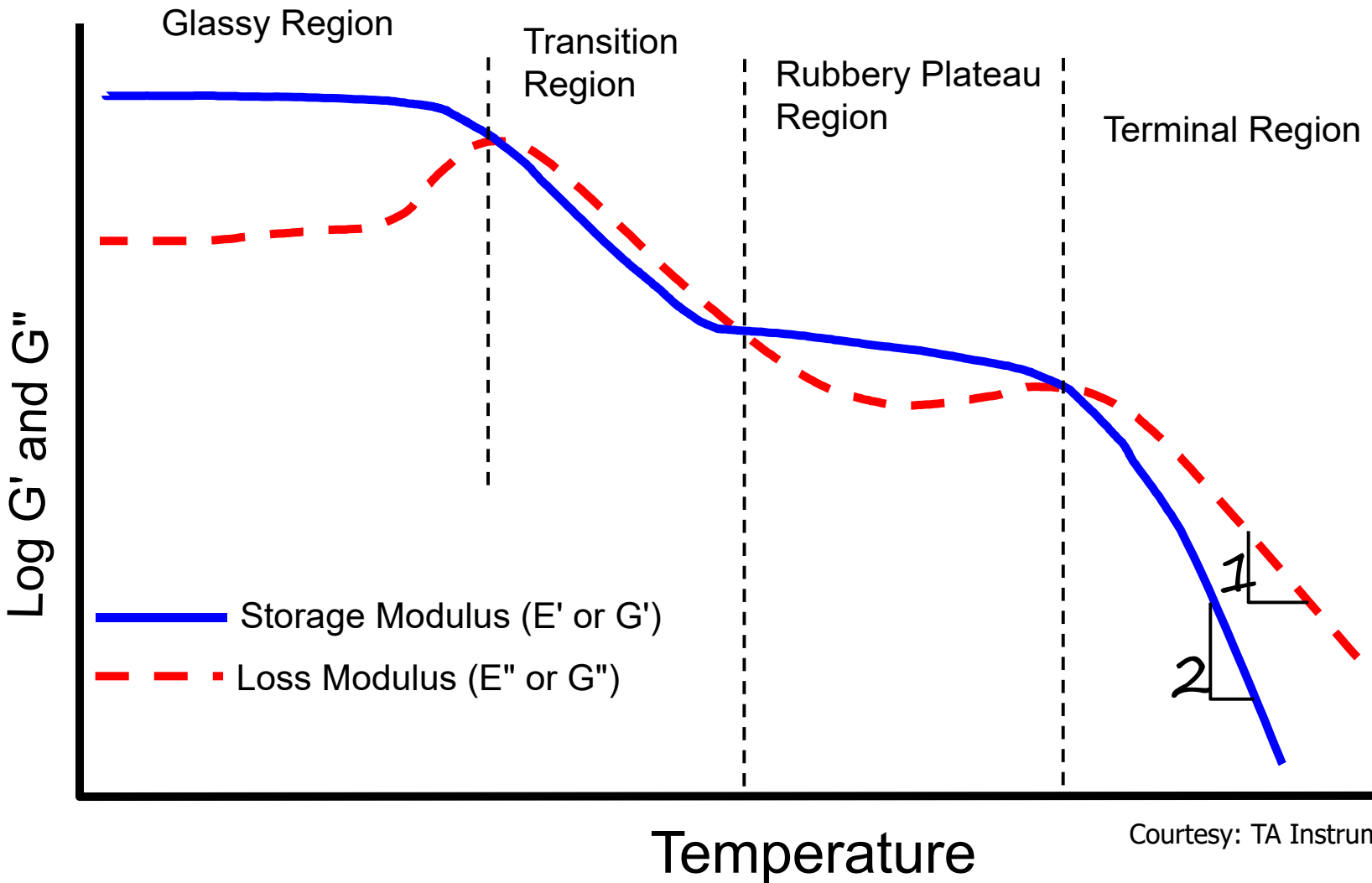
•Transition Zone:

- The period of oscillation is becoming too short to allow for complete rearrangement of chain conformation. Enough mobility is present for substantial friction between chain segments.

•Glassy Zone:

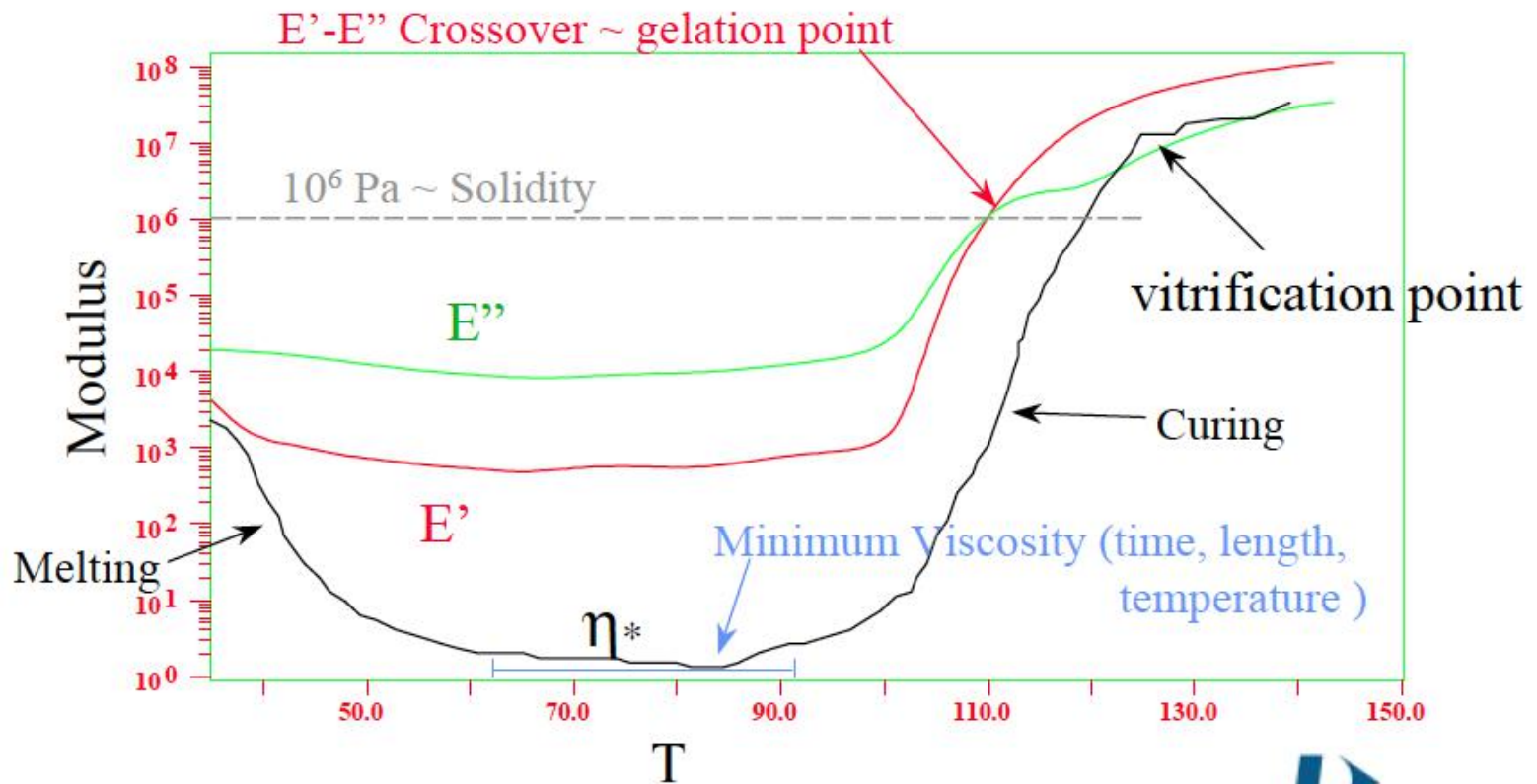
- No configurational rearrangements occur within the period of oscillation. Stress response to a given strain is high (glass-like solid) and $\tan\delta$ is on the order of 0.1

Dynamic Temperature Ramp or Step and Hold: Material Response



Courtesy: TA Instruments

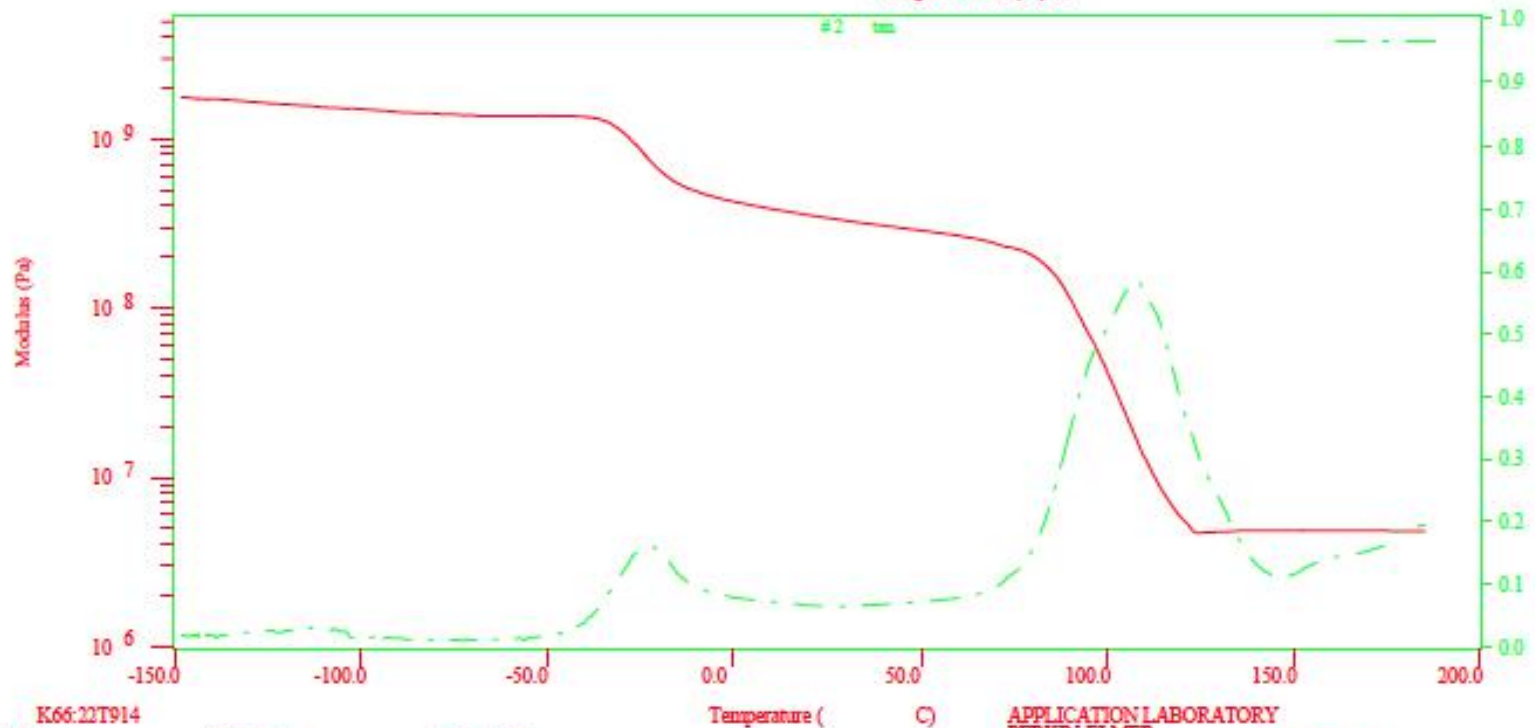
Analysis of a Cure by DMA



Blend

Curve 1: DMA Temp/Time Scan in 3 Point Bending
File info: str14 Thu Feb 15 10:45:19 1990
Frequency: 1.00 Hz Dynamic Stress: 2.00e+05Pa
STYRENE BUTADIENE RUBBER Tension: 110.000%

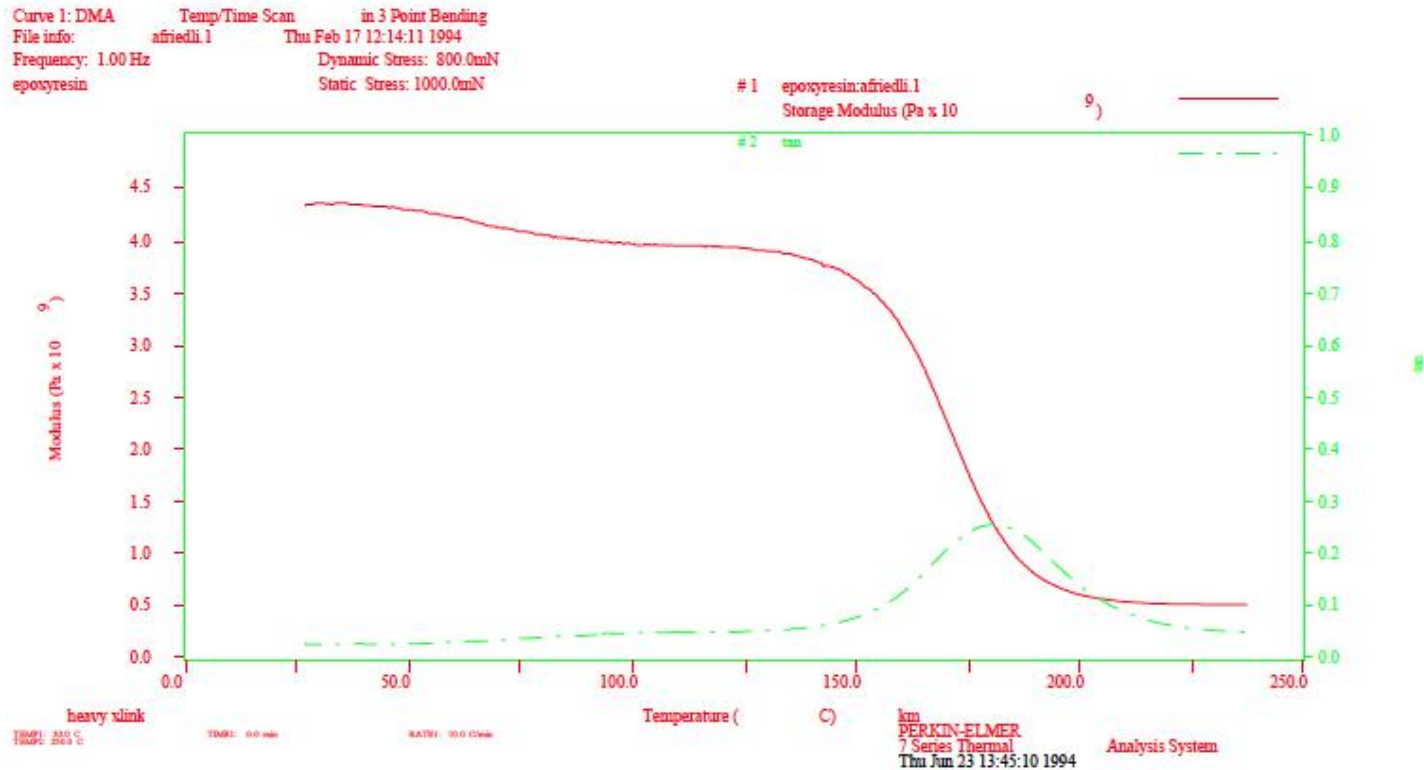
#1 STYRENE BUTADIENE RUBBER:str14
Storage Modulus (Pa) L



K66:22T914
T914P1: 180.0 C
T914P2: 350.0 C
T914P3: 0.0 min
RA101: 5.0 C/min

APPLICATION LABORATORY
PERKIN-ELMER
7 Series Thermal
Sun Nov 26 20:54:43 1995
Analysis System

Epoxy



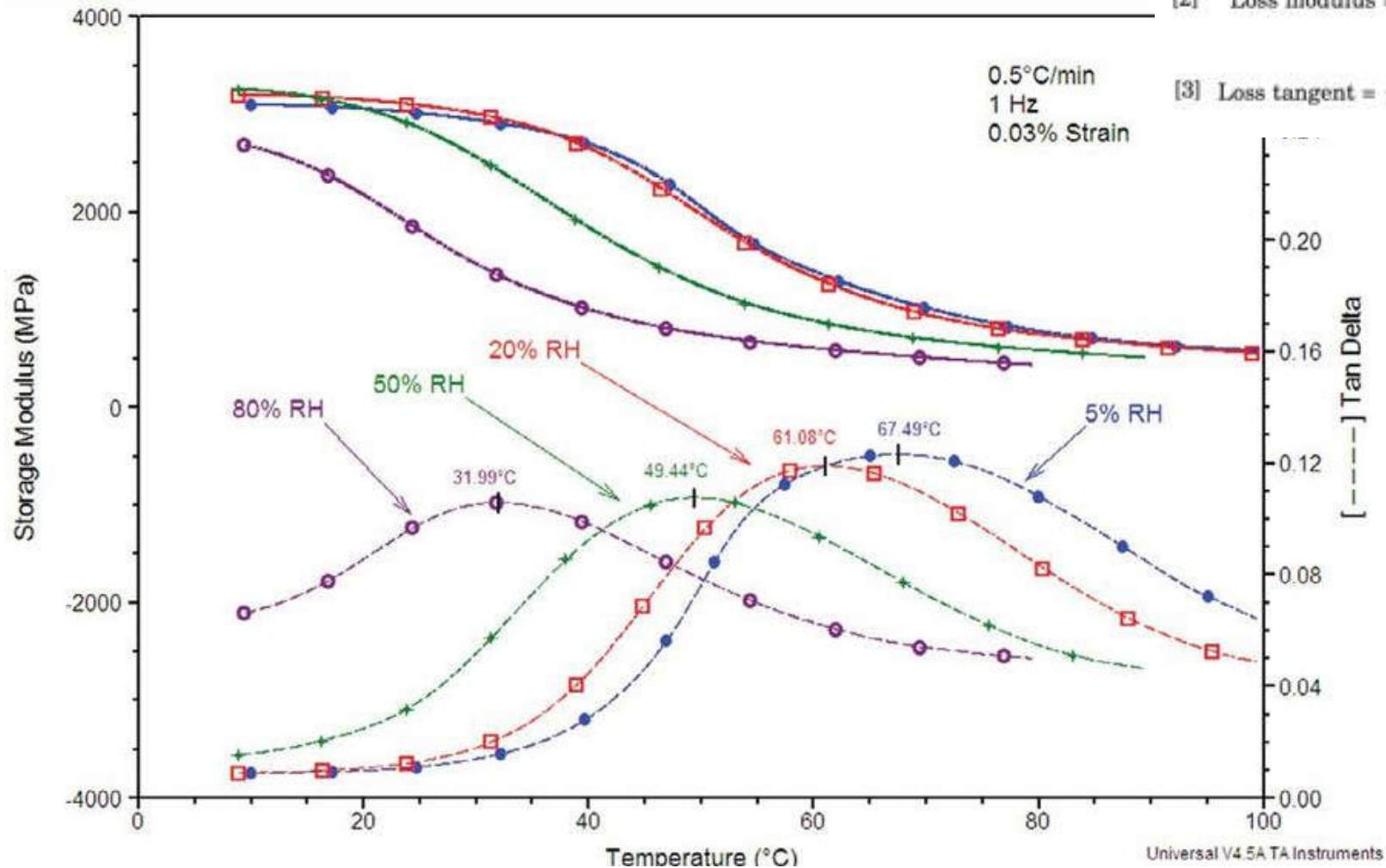
This T_g is undetectable in the DSC !!!!!!

Nylon-6 as a function of humidity

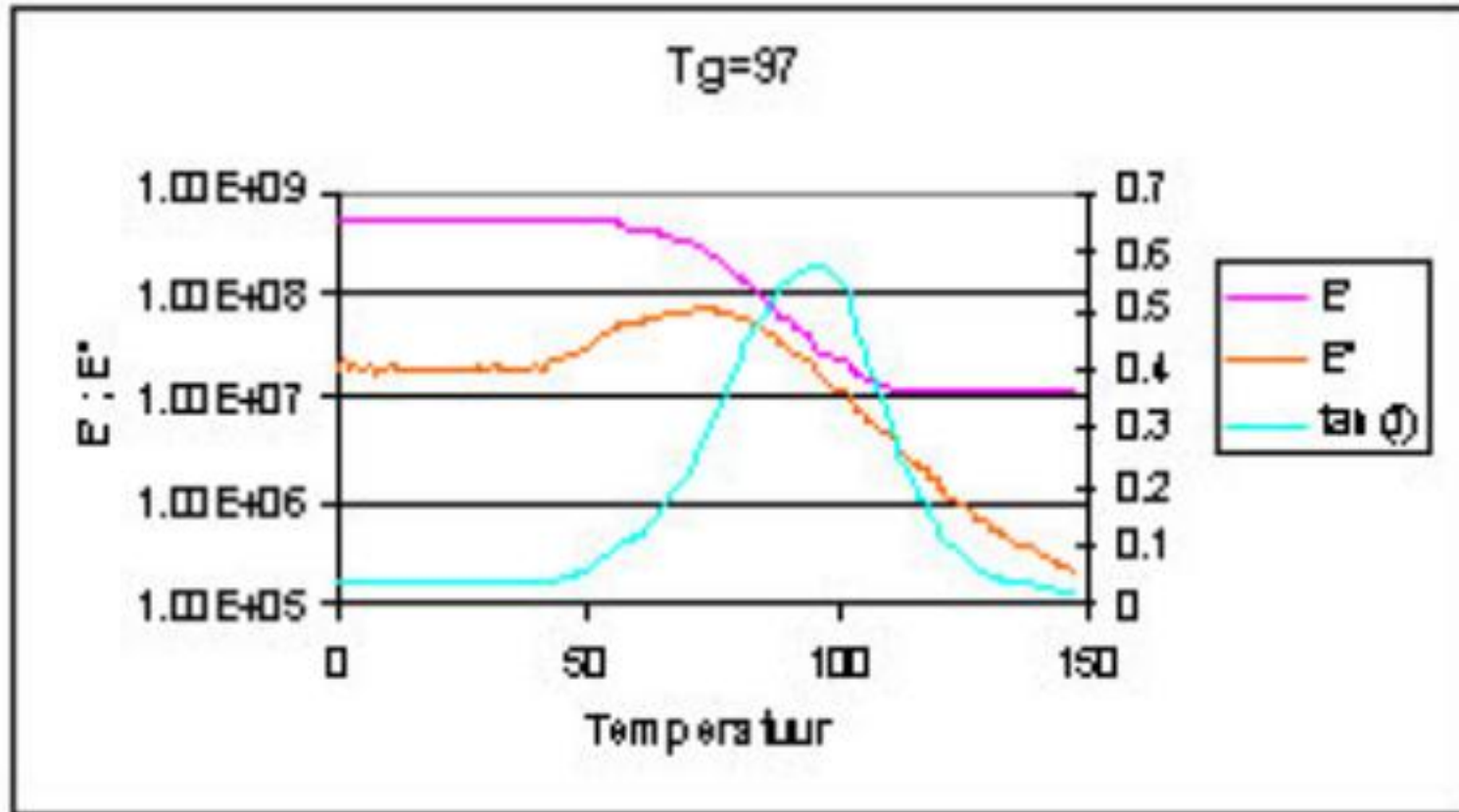
$$[1] \text{ Storage modulus} = \frac{\sigma_0 \cos \delta}{\epsilon_0} = E'$$

$$[2] \text{ Loss modulus} = \frac{\sigma_0 \sin \delta}{\epsilon_0} = E''$$

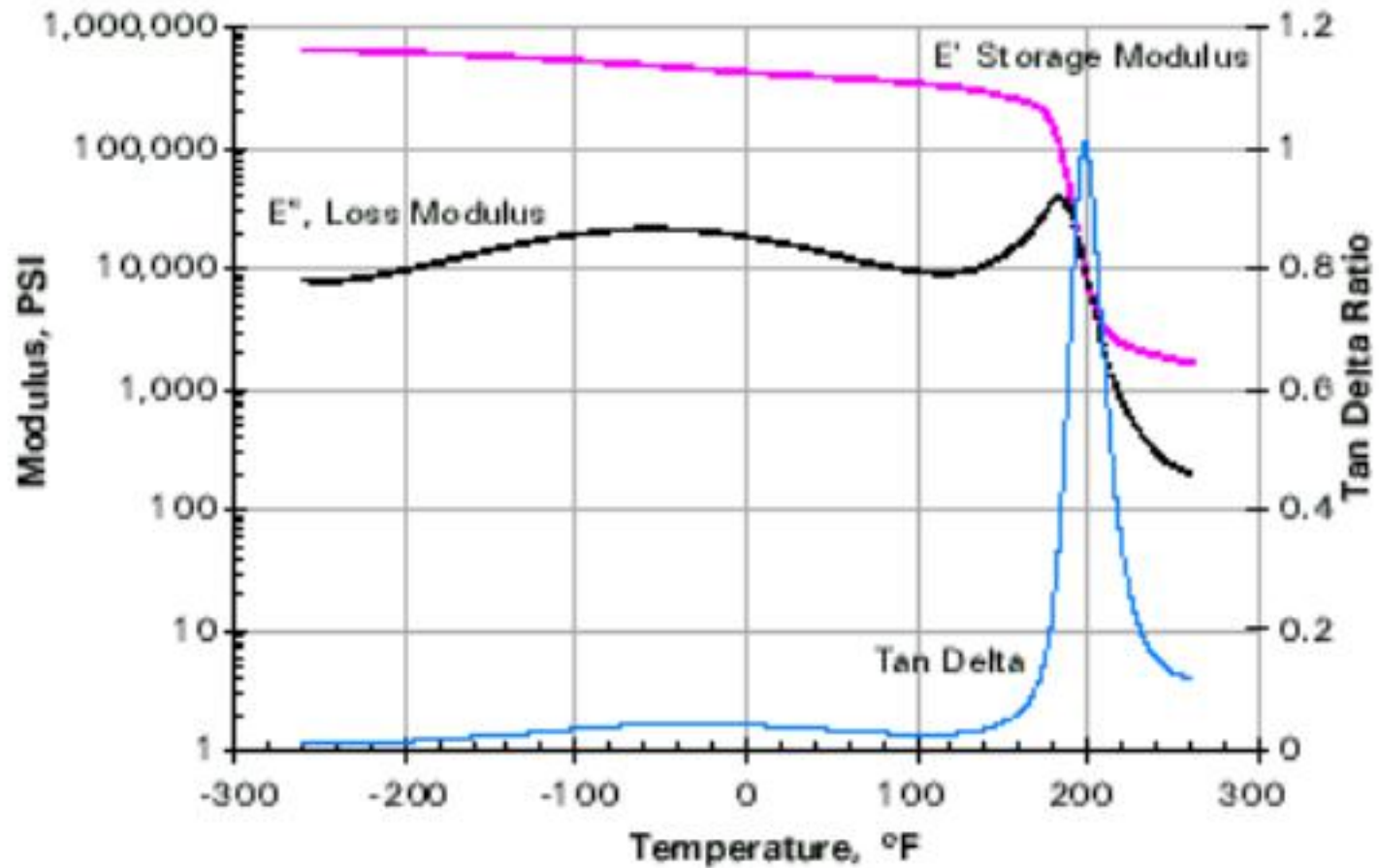
$$[3] \text{ Loss tangent} = \frac{E'}{E''} = \tan \delta$$



Poly(lactic acid)



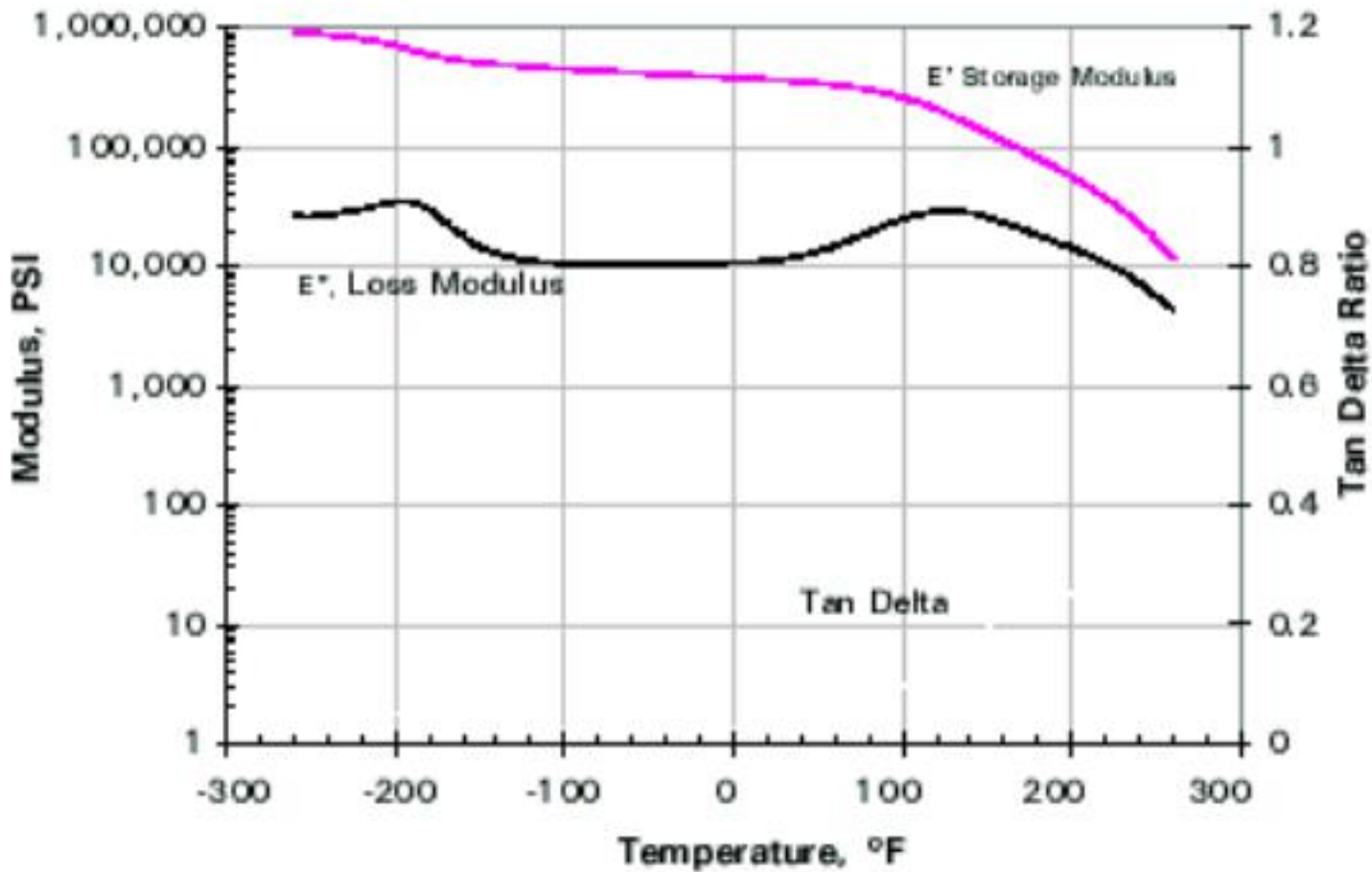
PVC Pipe Dynamic Mechanical Analysis



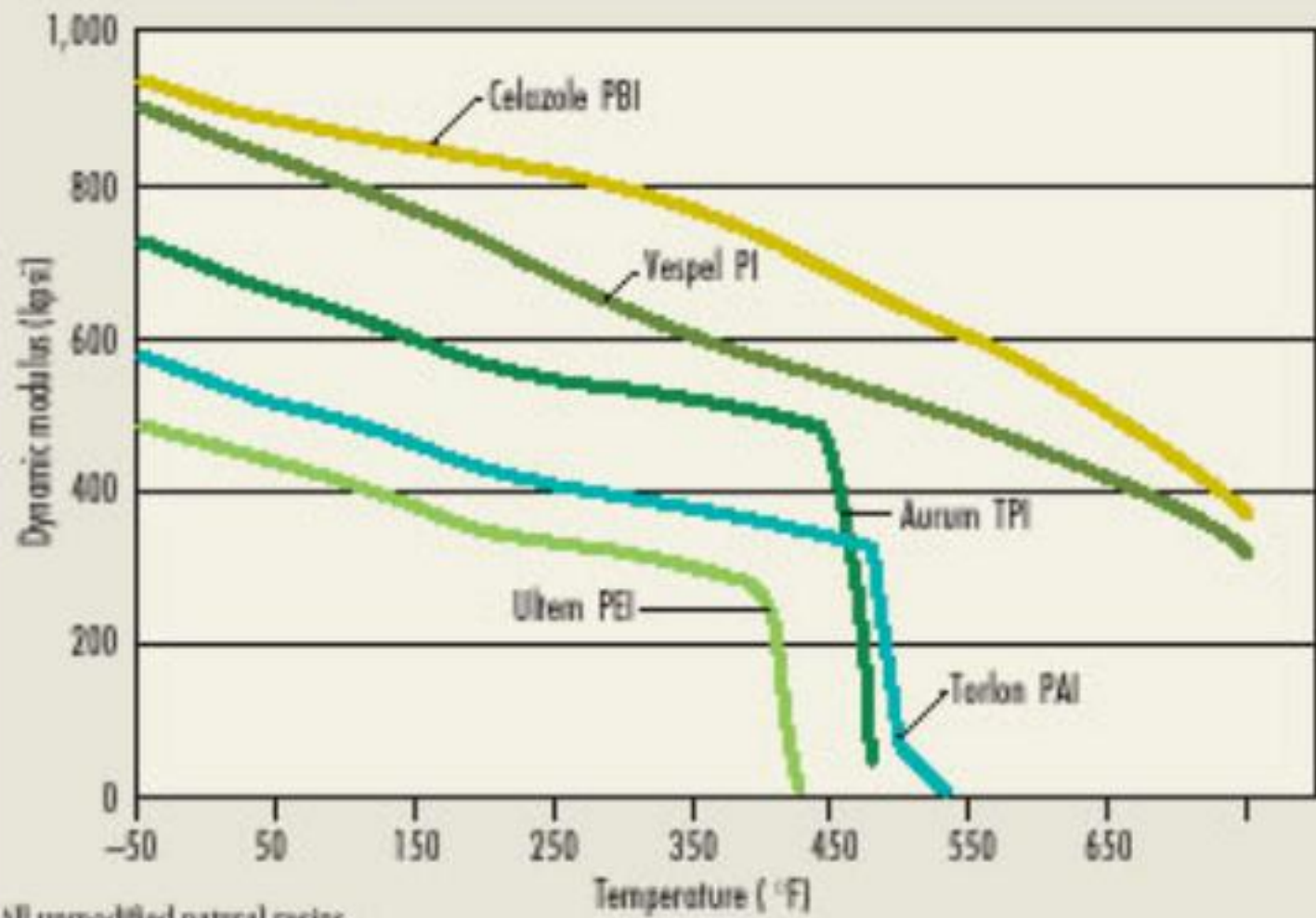
Tg 87 ° C

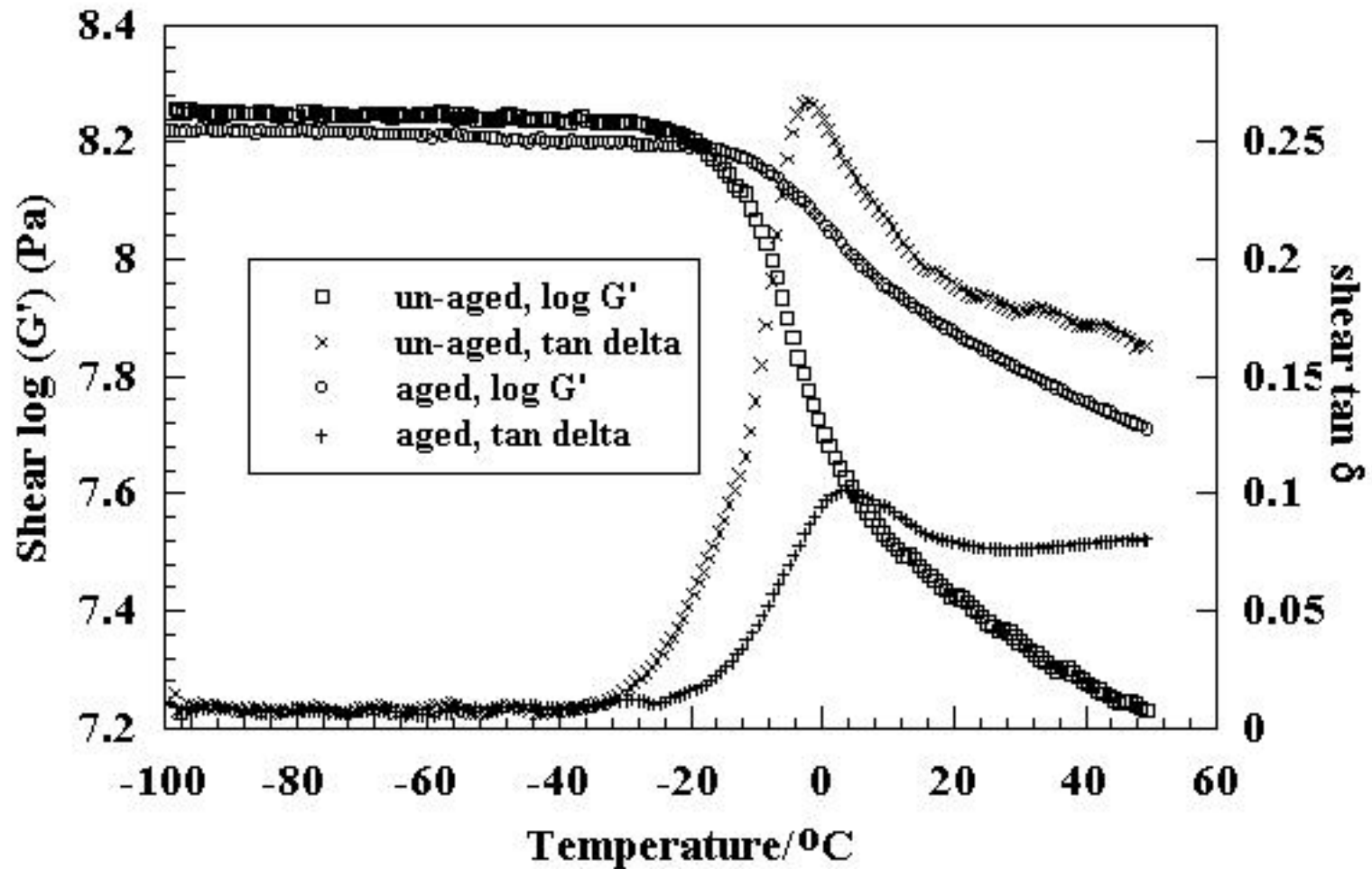
HDPE

Dynamic Mechanical Analysis



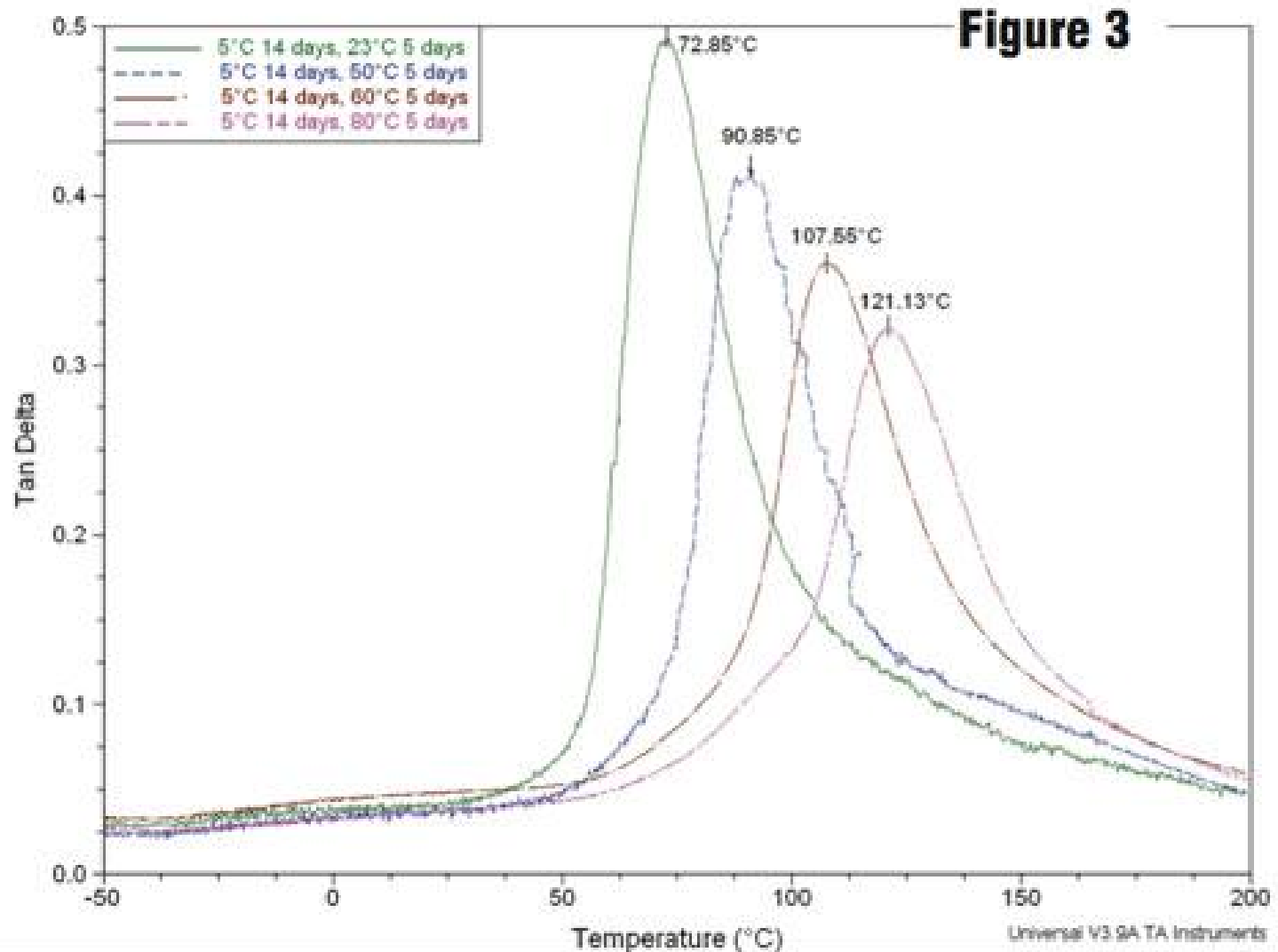
T_g -123°C (-190°F)
 T_m 135°C (275°F)



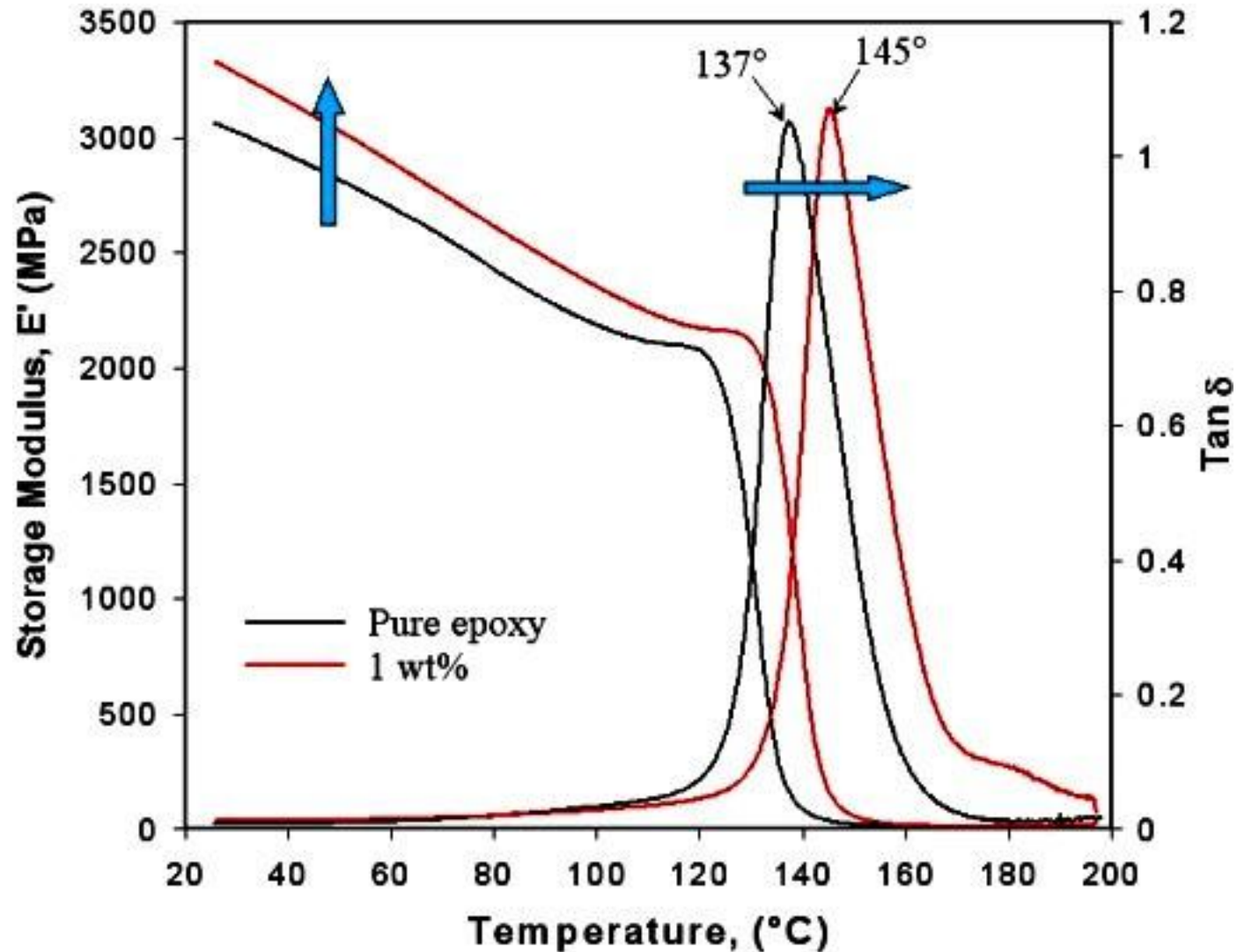


These data show the difference between the behaviour of un-aged and aged samples of rubber, and were collected in shear mode on the DMTA at 1 Hz. The aged sample has a lower modulus than the un-aged, and is weaker. The loss peak is also much smaller for the aged sample.

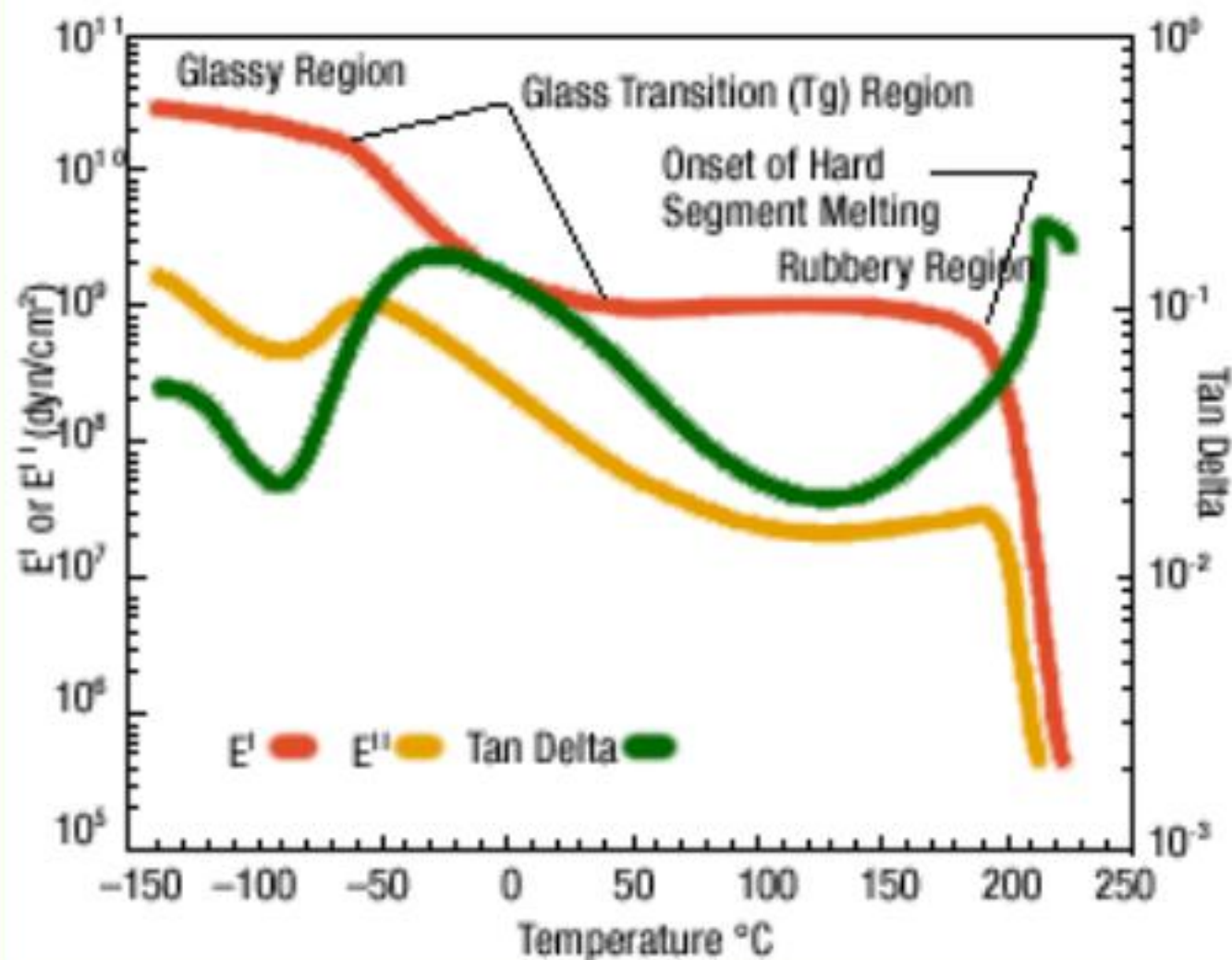
Tan δ of paint as it dries



Epoxy and epoxy with clay filler



Dynamic Mechanical Testing Data for a Typical Polyurethane Elastomer



HOMework

- Find, in reference(s), **DMA** figure(s) of polymer nanocomposites with carbon nanotubes and/or graphene, and explain the curves in details from what you have learnt.